# Chapter 5

# Pricing of Bermudan Style Libor Derivatives

## 5.1 Orientation

This chapter is devoted to the pricing of Bermudan style Libor derivatives by Monte Carlo methods. A Bermudan Libor derivative is an American option on a pay-off function of forward Libors, which may be called at a finite number of exercise dates. As such we have a high dimensional pricing problem due to the typically high dimension of the underlying Libor process. Generally, evaluation of American style derivatives on a high dimensional system of underlyings is considered a perennial problem for the last decades. On the one hand such high dimensional options are difficult, if not impossible, to compute by PDE methods for free boundary value problems. On the other hand Monte Carlo simulation, which is for high dimensional European options an almost canonical alternative to PDE solving, is for American options highly non-trivial since the (optimal) exercise boundary is usually unknown.

In the past literature, many approaches for Monte Carlo simulation of American options are developed. With respect to Bermudan derivatives, which are in fact American options with a finite number of exercise dates, there is, for example, the stochastic mesh method of Broadie & Glasserman (1997, 2000), a cross-sectional regression approach by Longstaff & Schwartz (2001), and a method by Andersen (1999) for computation of a parametric approximation of the exercise boundary. In general, the price of an American option can be represented as a supremum over a set of stopping times. As a remarkable result, Rogers (2001) (and independently Haugh & Kogan (2001) for Bermudan style instruments) showed that this supremum representation can be converted into a "dual" infimum representation, where the infimum is taken over a set of (super-)martingales. The idea behind this duality approach is rooted in earlier work of Davis & Karatzas (1994) in fact. In Joshi & Theis (2002) duality is used for finding Bermudan swaption prices via a minimization procedure. An alternative (multiplicative) dual approach based on change of numeraires is proposed by Jamshidian (2003,2004). Kolodko & Schoenmakers (2003,2004a) propose a trick for reducing the numerical complexity of computing upper bounds by duality and in Kolodko & Schoenmakers (2004) a general convergent procedure for the optimal Bermudan stopping time is developed. Further recent papers on methods for high-dimensional American options include Belomestny & Milstein (2004), Milstein, Reiß & Schoenmakers (2003), Berridge & Schumacher (2004), and for a more detailed and general overview we refer to Glasserman (2003) and the references therein.

In this chapter we will give a concise recap of the Bermudan pricing problem and describe different Monte Carlo methods for Bermudan style derivatives with applications to Bermudan swaptions. Besides the method of Andersen (1999) and the duality approach of Rogers (2001) and Haugh & Kogan (2001), we present two methods recently developed in Kolodko & Schoenmakers (2003,2004,2004a). The method of Kolodko & Schoenmakers (2003,2004a), exposed in Section 5.8, deals with the problem of efficient numerical evaluation of dual Bermudan upper bounds. A new generic iterative procedure for constructing the optimal Bermudan stopping time, hence the Snell envelope, is developed in Kolodko & Schoenmakers (2004) and here presented in Sections 5.4-5.7. This procedure basically improves a canonical exercise policy for Bermudan options which is already not far from optimal usually; *exercise as soon as the cash-flow dominates all the Europeans ahead.* As such the method has a flavour of "policy iteration" but is rather different from standard policy iteration as in Howard (1960) and Puterman (1994). In particular, the here proposed iterative method does not rely on explicit knowledge of respective transition kernels, is essentially dimension independent, and gives very good results with only a few iterations. These are important features since in various applications, for example, in the case of an underlying (high-dimensional) Libor process transition kernels are not available. Finally, in Section 5.9.1, we deal with multiple callable Bermudans and sketch how the iterative procedure for standard Bermudans can be carried over to such products. For more details we refer to Bender & Schoenmakers (2004). The study of multiple exercise options has recently gained more attraction; see also Carmona & Dayanik (2004), Carmona & Touzi(2004), and Meinshausen & Hambly (2004).

## 5.2 The Bermudan Pricing Problem

In this section we introduce Bermudan style Libor derivatives with respect to an underlying Libor process in the setting of Theorem 1.3. Let us consider a subset of tenor dates $\mathbb{T} := \{\mathcal{T}_0, \mathcal{T}_1, \ldots, \mathcal{T}_k\} \subset \{0, T_1, \ldots, T_n\}$ with $0 = \mathcal{T}_0 < \mathcal{T}_1 < \mathcal{T}_2 < \cdots < \mathcal{T}_k \leq T_n$. An option issued at time $t = 0$, to exercise once a cash-flow $C_{\mathcal{T}_\tau} := B_n(\mathcal{T}_\tau)c(\mathcal{T}_\tau, L(\mathcal{T}_\tau))$, for some given function $c(\cdot, \cdot)$, at a date $\mathcal{T}_\tau \in \mathbb{T}$ to be decided by the option holder, is called a Bermudan style Libor derivative. We consider the Libor dynamics with respect to a certain pricing measure $P$ connected with some discounting numeraire $B$, where $B$ is such that $B_n(t)/B(t)$ is measurable with respect to (i.e., is determined by) $L(s)$, $0 \leq s \leq t$. For example, we can take the dynamics (2.1) in the spot Libor measure where $B$ is the spot Libor rolling-over account (1.20), or SDE (1.16) in the terminal measure where $B$ is taken to be the terminal bond $B_n$. The value of the Bermudan derivative at time $t$, $t \geq 0$ (when the option is not exercised before $t$), is then given by

$$V(t) = B(t) \sup_{\tau \in \{\kappa(t),\ldots,k\}} E_P^{\mathcal{F}_t} \frac{C_{\mathcal{T}_\tau}}{B(\mathcal{T}_\tau)} =: B(t) \sup_{\tau \in \{\kappa(t),\ldots,k\}} E^{\mathcal{F}_t} Z^{(\tau)} \qquad (5.1)$$

with $\kappa(t) := \min\{m : \mathcal{T}_m \geq t\}$, and $Z^{(\tau)}$ denoting the discounted cash-flow at $\mathcal{T}_\tau$. The supremum in (5.1) is taken over all integer valued $\mathbb{F}$-stopping times $\tau$ with values in the set $\{\kappa(t), ..., k\}$, where $\mathbb{F} := \{\mathcal{F}_t,\ 0 \leq t \leq T\}$ denotes the usual filtration generated by the Libor process $L$ (see also Theorem 1.3). For technical reasons it is assumed that $Z^{(i)}$ has finite expectation for each $i$, $0 \leq i \leq k$. Note that $V(t)$ can also be seen as the price of a Bermudan option newly issued at time $t$, with exercise opportunities $\mathcal{T}_{\kappa(t)}, \ldots, \mathcal{T}_k$. The fact that (5.1) can be considered as the fair price for the Bermudan derivative is due to general no-arbitrage principles, e.g., see Duffie (2001).

A family of $\mathbb{F}$-stopping times (indices) $(\tau_t^*)_{0 \leq t \leq \mathcal{T}_k}$, with $\tau_t^* \in \{\kappa(t), ..., k\}$, is called an *optimal stopping family*, if

$$Y^*(t) := \frac{V(t)}{B(t)} = \sup_{\tau \in \{\kappa(t),\ldots,k\}}, E^{\mathcal{F}_t} Z^{(\tau)} = E^{\mathcal{F}_t} Z^{(\tau_t^*)}, \quad 0 \leq t \leq \mathcal{T}_k.$$

The thus induced process $Y^*$, called the *Snell envelope process*, is a supermartingale. Indeed, let $(\tau_t^*)$ be an optimal stopping family (which exists by general arguments) and $s < t$. Then it holds

$$E^{\mathcal{F}_s} Y^*(t) = E^{\mathcal{F}_s} E^{\mathcal{F}_t} Z^{(\tau_t^*)} = E^{\mathcal{F}_s} Z^{(\tau_t^*)} \leq \sup_{\tau \in \{\kappa(s),\ldots,k\}} E^{\mathcal{F}_s} Z^{(\tau)} = Y^*(s).$$

Henceforth we will consider the process $Y^*$ in particular at the exercise dates $\{\mathcal{T}_0, \mathcal{T}_1, ..., \mathcal{T}_k\}$, and we therefore introduce the discrete process $Y^{*(j)} := Y^*(\mathcal{T}_j)$, $0 \leq j \leq k$, together with the discrete filtration $\mathcal{F}^{(j)} := \mathcal{F}_{\mathcal{T}_j}$, $0 \leq j \leq k$. Further we consider a corresponding discrete stopping family $(\tau_j^*)$, where $\tau_j^* := \tau_{\mathcal{T}_j}^*$, for $0 \leq j \leq k$.

## Backward Dynamic Programming

It is well known and intuitively obvious that the discrete Snell envelope

$$Y^{*(j)} = E^{\mathcal{F}^{(j)}} Z^{(\tau_j^*)}, \qquad 0 \leq j \leq k, \qquad (5.2)$$

can be constructed by the following algorithm, called *backward dynamic programming* (see, e.g., Shiryayev (1978), Elliot & Kopp, 1999). At the last exercise date we have $Y^{*(k)} = Z^{(k)}$ and for $0 \le j < k$,

$$Y^{*(j)} = \max\left(Z^{(j)}, E^{\mathcal{F}^{(j)}} Y^{*(j+1)}\right). \tag{5.3}$$

An optimal stopping family is next represented by

$$\tau_i^* = \inf\left\{j, i \le j \le k : Y^{*(j)} \le Z^{(j)}\right\}. \tag{5.4}$$

Consider a situation where the discrete process $j \to (L(T_j), B(T_j))$ is Markovian. For instance, consider a Libor market model in the spot Libor numeraire or in the terminal bond measure. Then, obviously, $Y^{*(j)}$ is a function of $L(\mathcal{T}_j)$ and $B(\mathcal{T}_j)$ for $0 \le j \le k$, and thus the conditional expectations in (5.3) may be simulated by Monte Carlo using (regular) conditional probabilities $P^{(L(\mathcal{T}_j),B(\mathcal{T}_j))}$ (see Appendix A.1.4). So it is possible, in principle, to construct the optimal stopping time via (5.4) while the Snell envelope is constructed backwardly by straightforward Monte Carlo simulation of the backward dynamic program (5.3). But, the complexity of this procedure would be formidable due to its nested structure: $N$ simulations for computing each conditional expectation in (5.3) would lead to $O(N^k)$ samples for the whole set of $k$ exercise dates. For example, $N = 40$ and $k = 10$ would lead to about $40^{10}$ ($> 10^{16}$) samples!

**REMARK 5.1** It is always possible to choose the numeraire $B$ such that $B(T_j)$ is a function of $L(T_j)$ for all $j$, $0 \le j \le m(\mathcal{T}_k)$, with $m(\cdot)$ as in (1.20). For instance, one could take for $B$ simply the terminal bond $B_n$, or any linear combination of $B_{m(\mathcal{T}_k)}, \ldots, B_n$. For a Markovian Libor model (e.g., a market model) with respect to such a numeraire the conditional expectations in (5.2) may then be replaced with $E^{L(\mathcal{T}_j)}$ rather than $E^{(L(\mathcal{T}_j),B(\mathcal{T}_j))}$, $0 \le j \le k$, as in the case of the spot Libor numeraire. For notational convenience we will therefore assume such kind of numeraires in the description of several Monte Carlo methods in this chapter. It will be obvious, however, how to interpret the presented algorithms, for example, in terms of the spot Libor numeraire which may be more suitable for certain products in practice. ∎

## 5.3 Backward Construction of the Exercise Boundary

As a first practically feasible procedure for pricing Bermudan derivatives, we discuss a method for constructing the exercise region of a Bermudan product via backward induction. Obviously, once the exercise region is known, the Bermudan price can be computed by straightforward Monte Carlo simulation.

Let $\mathcal{B}^{(l)}$ be a Bermudan option which is only exercisable at $\mathcal{T}_l, ..., \mathcal{T}_k$, and $\tau_l^*$ be an optimal stopping time for $\mathcal{B}^{(l)}$, for $0 \leq l \leq k$. If $V^{(l)}(0)$ is the value of $\mathcal{B}^{(l)}$ at $t = 0$, we have

$$\begin{aligned}\frac{V^{(l)}(0)}{B(0)} &= E^{\mathcal{F}_0} Y^{*(l)} = E^{\mathcal{F}_0}\left(Y^{*(l)} 1_{\tau_l^* > l} + Z^{(l)} 1_{\tau_l^* = l}\right) \\ &= E^{\mathcal{F}_0}\left(1_{\tau_l^* > l} E^{\mathcal{F}^{(l)}} Z^{(\tau_l^*)}\right) + E^{\mathcal{F}_0}\left(Z^{(l)} 1_{\tau_l^* = l}\right) \\ &= E^{\mathcal{F}_0}\left(1_{\tau_l^* > l} E^{\mathcal{F}^{(l)}} Z^{(\tau_{l+1}^*)}\right) + E^{\mathcal{F}_0}\left(Z^{(l)} 1_{\tau_l^* = l}\right) \\ &= E^{\mathcal{F}_0}\left(1_{\tau_l^* > l} Z^{(\tau_{l+1}^*)}\right) + E^{\mathcal{F}_0}\left(Z^{(l)} 1_{\tau_l^* = l}\right). \qquad (5.5)\end{aligned}$$

We now assume a Markovian Libor model (e.g., a market model) and the numeraire $B$ to be as in Remark 5.1. For instance, think of $B$ being the terminal bond $B_n$. Then, due to Markovianity of $L$, we may replace in (5.5) $\mathcal{F}^{(l)}$ by the sigma algebra generated by $L(T_l)$, and by general arguments there exists for each $l$, $0 \leq l \leq k$, a smooth function $H_*^{(l)} : R^{n-1} \to R$ which describes the optimal exercise region for the Bermudan $\mathcal{B}^{(l)}$ at $\mathcal{T}_l$ by $H_*^{(l)}(L(\mathcal{T}_l)) \geq 0$. As a consequence,

$$\tau_l^* = \inf\{j : j \geq l,\ H_*^{(j)}(L(\mathcal{T}_j)) \geq 0\}, \qquad (5.6)$$

so $\{\tau_l^* = l\} = \{H_*^{(l)}(L(\mathcal{T}_l)) \geq 0\}$. Since (without loss of generality) cash-flows are assumed to be non-negative, we have $\tau_k^* \equiv k$ and thus we may take, for instance,

$$H_*^{(k)}(L(\mathcal{T}_k)) \equiv 0. \qquad (5.7)$$

Hence $H_*^{(k)}$ is considered to be trivially known. Equation (5.5) may be expressed in terms of the $H_*^{(l)}$ by

$$\frac{V^{(l)}(0)}{B(0)} = E^{\mathcal{F}_0}\left(1_{H_*^{(l)}(L(\mathcal{T}_l)) < 0} Z^{(\tau_{l+1}^*)}\right) + E^{\mathcal{F}_0}\left(Z^{(l)} 1_{H_*^{(l)}(L(\mathcal{T}_l)) \geq 0}\right). \qquad (5.8)$$

Suppose that for some $l$, $0 \leq l < k$, the functions $H_*^{(j)}$, $l < j \leq k$, are known (which is by (5.7) the case for $l = k - 1$). Then, the stopping rule $\tau_{l+1}^*$ is known by (5.6), and $H_*^{(l)}$ is (formally) a solution of the following optimization problem,

$$\frac{V^{(l)}(0)}{B(0)} = \max_{H \in \mathcal{H}} E^{\mathcal{F}_0}\left(1_{H(L(\mathcal{T}_l)) < 0} Z^{(\tau_{l+1}^*)}\right) + E^{\mathcal{F}_0}\left(Z^{(l)} 1_{H(L(\mathcal{T}_l)) \geq 0}\right), \qquad (5.9)$$

where $\mathcal{H}$ is some suitable class of smooth functions $H : R^{n-1} \to R$. Based on some procedure for the optimization problem (5.9), we may proceed, at least in principle, to construct the whole sequence $H_*^{(l)}$, $0 \leq l < k$, backwardly. This general procedure is utilized by Andersen (1999) in the context of Bermudan swaptions in the Libor market model.

### Andersen's Method

In fact, the method of Andersen (1999) can be applied to general Bermudan style derivatives and essentially works as follows. Let us assume that $H_*^{(j)}$ is known for $l < j \leq k$. By (5.7) this is always the case for $l = k-1$. Consider some rich enough parametric family of functions $H_\alpha^{(l)} : R^{n-1} \to R$, where $\alpha$ runs through some parameter set $A$. For example, $A \subset R^q$ for some integer $q$. Then, clearly,

$$\frac{V^{(l)}(0)}{B(0)} \geq \sup_{\alpha \in A} \left( E^{\mathcal{F}_0} \left( 1_{H_\alpha^{(l)}(L(\mathcal{T}_l))<0} Z^{(\tau_{l+1}^*)} \right) + E^{\mathcal{F}_0} \left( Z^{(l)} 1_{H_\alpha^{(l)}(L(\mathcal{T}_l))\geq 0} \right) \right),$$

since each particular $\alpha$ represents a sub-optimal exercise decision. Now consider a Monte Carlo simulation of sample paths $L^{(m)}$, $m = 1, ..., M$, with initial value $L(0)$. Since $H_*^{(j)}$, $l < j \leq k$, is known by assumption, we know the stopping rule $\tau_{l+1}^*$ by (5.6), and then we may carry out the optimization

$$\frac{1}{M} \sum_{m=1}^{M} \left( 1_{H_\alpha^{(l)}(L^{(m)}(\mathcal{T}_l))<0} Z_m^{(\tau_{l+1}^*)} + Z_m^{(l)} 1_{H_\alpha^{(l)}(L^{(m)}(\mathcal{T}_l))\geq 0} \right) \to \sup_{\alpha \in A},$$

where $Z_m^{(l)} := Z^{(l)}(L^{(m)}(\mathcal{T}_l))$. If $\alpha_l^* \in A$ is a parameter where the supremum is attained, the family $H_{\alpha_l^*}^{(l)}, H_*^{(l+1)}, \ldots, H_*^{(k)}$ provides a backward extension of the exercise boundary to the exercise dates $\mathcal{T}_l, ..., \mathcal{T}_k$. In the ideal situation the parametric family $\{H_\alpha^{(l)}, \alpha \in A\}$ contains the optimal $H_*^{(l)}$ and then, naturally, $H_{\alpha_l^*}^{(l)} \to H_*^{(l)}$ (in a certain sense) as $M \to \infty$. In practice, however, it is usually not possible to identify such a family, and $M$ can not be arbitrarily large. But, it may be possible to find a simple family which admits a close approximation of the optimal strategy, and then, for large enough $M$, the sequence $H_{\alpha_l^*}^{(l)}, H_*^{(l+1)}, \ldots, H_*^{(k)}$ may be considered to give a good approximation for the exercise boundary at the exercise dates $\mathcal{T}_l, ..., \mathcal{T}_k$. Next, we proceed to compute $H_{\alpha_{l-1}^*}^{(l-1)}$, down to $H_{\alpha_0^*}^{(0)}$ in the same way. The backward construction starts with $l = k-1$. As a most important feature, *one and the same set of sample paths can be used for each recursion step.*

### Bermudan Swaptions

As an example, we consider a Bermudan payer swaption with strike $\theta$ on the underlying swaps $S_{\mathcal{T}_j,T_n}$,[1] $0 \leq j \leq k$, with swap starting date $\mathcal{T}_j$ and fixed swap end $T_n$. The Bermudan discounted cash-flows at the exercise tenors are then given by

$$Z^{(j)} := \frac{B_{\mathcal{T}_j,T_n}(S_{\mathcal{T}_j,T_n}(\mathcal{T}_j) - \theta)^+}{B(\mathcal{T}_j)}, \quad 0 \leq j \leq k, \tag{5.10}$$

[1] For obvious reasons, we here choose a notation for swap rate and annuity numeraire which differs slightly from the respective notations in Chapters 1-4.

where $B_{T_p,T_q}$ is the annuity numeraire; see (1.24) or (1.26). In the present notations, Andersen (1999) proposed the following strategies.

For parameters $\alpha := (\alpha^{(0)}, ..., \alpha^{(k-1)})^T \in R^{k-1}$, $0 \leq j < k$ (note that we set $H_*^{(k)} \equiv 0$), we may take

*Exercise strategy I:*

$$H_\alpha^{(j)}(L(\mathcal{T}_j)) = B(\mathcal{T}_j)Z^{(j)} - \alpha^{(j)};$$

*Exercise strategy II:*

$$H_\alpha^{(j)}(L(\mathcal{T}_j)) = \min(B(\mathcal{T}_j)Z^{(j)} - \alpha^{(j)},\, Z^{(j)} - \max_{j<l\leq k} E^{\mathcal{F}^{(j)}} Z^{(l)});$$

*Exercise strategy III:*

$$H_\alpha^{(j)}(L(\mathcal{T}_j)) = B(\mathcal{T}_j)Z^{(j)} - \max_{j<l\leq k} B(\mathcal{T}_j)E^{\mathcal{F}^{(j)}} Z^{(l)} - \alpha^{(j)}.$$

Remarkably, as we will see from our experimental results in Section 5.8.3 and also reported in Andersen (1999) and Andersen & Broadie (2001), the simple strategy I works pretty well in practice.

## 5.4 Iterative Construction of the Optimal Stopping Time

As an alternative to the backward construction of the exercise boundary in the previous section we now discuss the iterative procedure for approximating the Snell envelope of Kolodko & Schoenmakers (2004). We first present a procedure which improves upon a given exercise policy for the Bermudan problem (5.1), represented by a family of stopping times.

### 5.4.1 A One Step Improvement upon a Given Family of Stopping Times

With respect to the discrete filtration $\left(\mathcal{F}^{(j)}\right)_{0\leq j\leq k}$ we consider some given family of integer valued stopping indexes $(\tau_i)$, with the following properties,

$$\begin{aligned} &i \leq \tau_i \leq k,\ \tau_k = k, \\ &\tau_i > i \Rightarrow \tau_i = \tau_{i+1}, \qquad 0 \leq i < k, \end{aligned} \tag{5.11}$$

and the process

$$Y^{(i)} := E^{\mathcal{F}^{(i)}} Z^{(\tau_i)}. \tag{5.12}$$

For example one could take

$$\tau_i := k \wedge \inf\{j \geq i : (\mathcal{T}_j, L(\mathcal{T}_j)) \in R\},$$

where $R$ is a certain region in $\mathbb{R}^n$, or, as a more trivial example, the family $\tau_i \equiv i$. Generally, the process $(Y^{(i)})$ is a lower approximation of the Snell envelope process $(Y^{*(i)})$ due to the family of (sub-optimal) stopping times $(\tau_i)$.

Based on the family $(\tau_i)$ we are going to construct a new family $(\widehat{\tau}_i)$ satisfying (5.11), which induces a new approximation of the Snell envelope. We first introduce an intermediate process

$$\widetilde{Y}^{(i)} := \max_{p:\, i \leq p \leq k} E^{\mathcal{F}^{(i)}} Z^{(\tau_p)}. \tag{5.13}$$

Using $\widetilde{Y}^{(i)}$ as a new exercise criterion we define a next family of stopping indexes

$$\begin{aligned} \widehat{\tau}_i : &= \inf\{j : i \leq j \leq k,\ \widetilde{Y}^{(j)} \leq Z^{(j)}\} \qquad (5.14)\\ &= \inf\{j : i \leq j \leq k,\ \max_{p:\, j \leq p \leq k} E^{\mathcal{F}^{(j)}} Z^{(\tau_p)} \leq Z^{(j)}\}, \quad 0 \leq i \leq k, \end{aligned}$$

and consider the process

$$\widehat{Y}^{(i)} := E^{\mathcal{F}^{(i)}} Z^{\widehat{\tau}_i} \tag{5.15}$$

as a next approximation of the Snell envelope. Clearly, the family $(\widehat{\tau})$ satisfies the properties (5.11) as well. For instance, the trivial family $\tau_i \equiv i$ gives for $\widetilde{Y}$ the maximum of still alive Europeans and for $\widehat{Y}$ the Bermudan price due to the stopping rule "wait until the first exercise date when the cash-flow is at least equal to the maximum of still alive Europeans". See for a more detailed explanation of these examples Section 5.8.2. As another example, $\tau_i \equiv k$ gives for $\widetilde{Y}$ the European option process due to the last exercise date $k$ and

$$\widehat{\tau}_i := \inf\{j : i \leq j \leq k,\ E^{\mathcal{F}^{(j)}} Z^{(k)} \leq Z^{(j)}\}, \quad 0 \leq i \leq k.$$

By the next crucial theorem, $(\widehat{Y}^{(i)})$ is generally an improvement of $(Y^{(i)})$.

***THEOREM 5.1***

*(Kolodko & Schoenmakers (2004)) Let $(\tau_i)$ be a family of stopping times with the property* (5.11) *and $(Y^{(i)})$ be given by* (5.12). *Let the processes $(\widetilde{Y}^{(i)})$ and $(\widehat{Y}^{(i)})$ be defined by* (5.13) and (5.15), *respectively. Then, it holds*

$$Y^{(i)} \leq \widetilde{Y}^{(i)} \leq \widehat{Y}^{(i)} \leq Y^{*(i)}, \quad 0 \leq i \leq k.$$

**PROOF**

The inequalities $Y^{(i)} \leq \widetilde{Y}^{(i)}$ and $\widehat{Y}^{(i)} \leq Y^{*(i)}$ are trivial. We only need to show the middle inequality. We use induction in $i$. Due to the definition of $\widetilde{Y}$ and $\widehat{Y}$, we have $\widetilde{Y}^{(k)} = \widehat{Y}^{(k)} = Z^{(k)}$. Suppose that $\widetilde{Y}^{(i)} \leq \widehat{Y}^{(i)}$ for some $i$ with $0 < i \leq k$. We will then show that $\widetilde{Y}^{(i-1)} \leq \widehat{Y}^{(i-1)}$. Let us write

$$\begin{aligned}\widehat{Y}^{(i-1)} &= E^{\mathcal{F}^{(i-1)}} Z^{(\widehat{\tau}_{i-1})} = 1_{\widehat{\tau}_{i-1}=i-1} Z^{(i-1)} + 1_{\widehat{\tau}_{i-1}>i-1} E^{\mathcal{F}^{(i-1)}} E^{\mathcal{F}^{(i)}} Z^{(\widehat{\tau}_i)} \\ &= 1_{\widehat{\tau}_{i-1}=i-1} Z^{(i-1)} + 1_{\widehat{\tau}_{i-1}>i-1} E^{\mathcal{F}^{(i-1)}} \widehat{Y}^{(i)}.\end{aligned}$$

Then, by induction,

$$\begin{aligned}\widehat{Y}^{(i-1)} &\geq 1_{\widehat{\tau}_{i-1}=i-1} Z^{(i-1)} + 1_{\widehat{\tau}_{i-1}>i-1} E^{\mathcal{F}^{(i-1)}} \widetilde{Y}^{(i)} \\ &= 1_{\widehat{\tau}_{i-1}=i-1} Z^{(i-1)} + 1_{\widehat{\tau}_{i-1}>i-1} E^{\mathcal{F}^{(i-1)}} \max_{p:\, i \leq p \leq k} E^{\mathcal{F}^{(i)}} Z^{(\tau_p)} \\ &\geq 1_{\widehat{\tau}_{i-1}=i-1} \widetilde{Y}^{(i-1)} + 1_{\widehat{\tau}_{i-1}>i-1} \max_{p:\, i \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}, \end{aligned} \tag{5.16}$$

since for $\widehat{\tau}_{i-1} = i-1$ we have $i-1 = \inf\{j : i-1 \leq j < k,\ \widetilde{Y}^{(j)} \leq Z^{(j)}\}$, and so

$$\widetilde{Y}^{(i-1)} = \max_{p:\, i-1 \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)} \leq Z^{(i-1)}.$$

We may write (5.16) as

$$\widehat{Y}^{(i-1)} \geq \widetilde{Y}^{(i-1)} + 1_{\widehat{\tau}_{i-1}>i-1} (\max_{i \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)} - \max_{i-1 \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}).$$

Thus, after showing that $\widehat{\tau}_{i-1} > i-1$ implies

$$E^{\mathcal{F}^{(i-1)}} Z^{(\tau_{i-1})} \leq \max_{p: i \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)},$$

it follows that $\widehat{Y}^{(i-1)} \geq \widetilde{Y}^{(i-1)}$. It holds,

$$\begin{aligned}E^{\mathcal{F}^{(i-1)}} Z^{(\tau_{i-1})} &= 1_{\tau_{i-1}=i-1} Z^{(i-1)} + 1_{\tau_{i-1}>i-1} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_i)} \\ &\leq 1_{\tau_{i-1}=i-1} Z^{(i-1)} + 1_{\tau_{i-1}>i-1} \max_{p: i \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}. \end{aligned} \tag{5.17}$$

Then, on the set $\widehat{\tau}_{i-1} > i-1$ we have

$$Z^{(i-1)} < \max_{p:\, i-1 \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)},$$

so if $(\widehat{\tau}_{i-1} > i-1) \wedge (\ \tau_{i-1} = i-1)$ it follows that

$$Z^{(i-1)} < \max(Z^{(i-1)}, \max_{p:\, i \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}).$$

Hence, if $(\widehat{\tau}_{i-1} > i-1) \wedge (\tau_{i-1} = i-1)$,

$$Z^{(i-1)} < \max_{p:\, i \leq p \leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}.$$

Thus, from (5.17) we have for $\widehat{\tau}_{i-1} > i-1$,

$$E^{\mathcal{F}^{(i-1)}} Z^{(\tau_{i-1})} \leq 1_{\tau_{i-1}=i-1} \max_{p:\, i\leq p\leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}$$
$$+1_{\tau_{i-1}>i-1} \max_{p:\, i\leq p\leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)} = \max_{p:\, i\leq p\leq k} E^{\mathcal{F}^{(i-1)}} Z^{(\tau_p)}.$$

∎

### 5.4.2 Iterating to the Optimal Stopping Time and the Snell Envelope

Naturally, we may construct by induction via the procedure (5.14)-(5.15) a sequence of pairs

$$\left((\tau_i^{(m)})_{0\leq i\leq k}, (Y^{m\,(i)})_{0\leq i\leq k}\right)_{m=0,1,2,\ldots}$$

in the following way: Start with some family of stopping times $(\tau_i^{(0)})_{0\leq i\leq k}$, which satisfies (5.11) and the additional requirement,

$$Y^{0\,(i)} := E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(0)})} \geq Z^{(i)}, \quad 0\leq i\leq k. \tag{5.18}$$

A canonical starting family is obtained, for example, by taking $\tau_i^{(0)} \equiv i$. Suppose that for $m\geq 0$ the pair $\left((\tau_i^{(m)}), (Y^{m\,(i)})\right)$ is constructed, where

$$Y^{m\,(i)} := E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m)})} \geq Z^{(i)}, \quad 0\leq i\leq k,$$

and the stopping time family $(\tau_i^{(m)})$ satisfies (5.11). Then define

$$\tau_i^{(m+1)} := \inf\{j : i\leq j\leq k,\ \max_{p:\, j\leq p\leq k} E^{\mathcal{F}^{(j)}} Z^{(\tau_p^{(m)})} \leq Z^{(j)}\}$$
$$= \inf\{j : i\leq j\leq k,\ \widetilde{Y}^{m+1\,(j)} \leq Z^{(j)}\}, \quad 0\leq i\leq k, \tag{5.19}$$

with

$$\widetilde{Y}^{m+1\,(i)} := \max_{p:\, i\leq p\leq k} E^{\mathcal{F}^{(i)}} Z^{(\tau_p^{(m)})}$$

being an intermediate dummy process and

$$Y^{m+1\,(i)} := E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m+1)})} \geq Z^{(i)}, \quad 0\leq i\leq k.$$

Clearly, $\tau_i^{(m+1)}$ satisfies (5.11) as well, and due to Theorem 5.1 we have

$$Z^{(i)} \leq Y^{0\,(i)} \leq Y^{m\,(i)} \leq \widetilde{Y}^{m+1\,(i)} \leq Y^{m+1\,(i)} \leq Y^{*\,(i)},\ 0\leq m<\infty,\ 0\leq i\leq k. \tag{5.20}$$

By the following proposition, for each fixed $i$ the sequence $(\tau_i^{(m)})_{m\geq 1}$ constructed above is non-decreasing in $m$ and bounded by any optimal stopping time $\tau_i^*$.

***PROPOSITION 5.1***

*Let $\tau_i^*$ be an optimal stopping time. For each $m$: $1 \leq m < \infty$ and $i$: $0 \leq i \leq k$, we have*

$$\tau_i^{(m)} \leq \tau_i^{(m+1)} \leq \tau_i^*.$$

**PROOF** Suppose that $\tau_i^* < \tau_i^{(m)}$ for some $m \geq 1$ and some $i$ with $0 \leq i \leq k$. Then, by (5.20) and the definition of $\tau_i^{(m)}$,

$$Y^{*(\tau_i^*)} \geq \widetilde{Y}^{m(\tau_i^*)} > Z^{(\tau_i^*)};$$

so $\tau_i^*$ is not optimal, hence a contradiction. Thus, the right inequality is proved. Next suppose $\tau_i^{(m+1)} < \tau_i^{(m)}$ for some $m \geq 1$ and some $i$ with $0 \leq i \leq k$. Then, by the definition of $\tau_i^{(m)}$ we have

$$\widetilde{Y}^{m(\tau_i^{(m+1)})} > Z^{(\tau_i^{(m+1)})}.$$

On the other hand, according to the definition of $\tau_i^{(m+1)}$, we have

$$\widetilde{Y}^{m+1(\tau_i^{(m+1)})} \leq Z^{(\tau_i^{(m+1)})}.$$

So, we get $\widetilde{Y}^{m(\tau_i^{(m+1)})} > \widetilde{Y}^{m+1(\tau_i^{(m+1)})}$, which contradicts (5.20). ∎

Due to the inequality chain (5.20) and Proposition 5.1 we now may define a limit lower bound process $Y^{\infty}$ and a limit family of stopping times $(\tau_i^{\infty})$ by

$$Y^{\infty(i)} := (\text{a.s.}) \lim_{m\uparrow\infty} \uparrow Y^{m(i)} \quad \text{and} \quad \tau_i^{\infty} := (\text{a.s.}) \lim_{m\uparrow\infty} \uparrow \tau_i^{(m)}, \quad 0 \leq i \leq k, \tag{5.21}$$

where the uparrows indicate that the respective sequences are non-decreasing. Further it is clear that the family $(\tau_i^{\infty})$ satisfies (5.11). Since $\tau_i^{(m)}$ is an integer valued random variable in the set $\{i, \ldots, k\}$ for each $m$, we have with probability 1, $\tau_i^{(m)}(\omega) = \tau_i^{\infty}(\omega)$ for $m > N(\omega)$. Therefore, $Z^{(\tau_i^{(m)})} \to Z^{(\tau_i^{\infty})}$ with probability 1, and so by dominated convergence we have

$$Y^{\infty(i)} = (\text{a.s.}) \lim_{m\uparrow\infty} \uparrow E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m)})} = E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{\infty})}, \quad 0 \leq i \leq k.$$

We are now ready to present our main result.

***THEOREM 5.2***

*(Kolodko & Schoenmakers (2004)) The constructed limit process $Y^{\infty}$ in (5.21) coincides with the Snell envelope process $Y^*$, and $(\tau_i^{\infty})$ in (5.21) acts as a*

*family of optimal stopping times. We have*

$$Y^{*(i)} = Y^{\infty(i)} = E^{\mathcal{F}^{(i)}} Z^{(\tau_i^\infty)}, \quad 0 \le i \le k. \tag{5.22}$$

**PROOF** The Snell envelope $\left(Y^{*(i)}\right)$ is the smallest supermartingale which dominates the process $Z$ (see, e.g., Shiryayev (1978), Elliot & Kopp, 1999). So, by (5.21) and the first inequality in (5.20) it is enough to show that the process $\left(Y^{\infty(i)}\right)$ is a supermartingale. Note, that for each $i$: $0 \le i < k$,

$$\begin{aligned} E^{\mathcal{F}^{(i)}} Y^{\infty(i+1)} &= E^{\mathcal{F}^{(i)}} E^{\mathcal{F}^{(i+1)}} Z^{(\tau_{i+1}^\infty)} \\ &= E^{\mathcal{F}^{(i)}} 1_{\tau_i^\infty = i} E^{\mathcal{F}^{(i+1)}} Z^{(\tau_{i+1}^\infty)} + E^{\mathcal{F}^{(i)}} 1_{\tau_i^\infty > i} E^{\mathcal{F}^{(i+1)}} Z^{(\tau_{i+1}^\infty)} \\ &= E^{\mathcal{F}^{(i)}} 1_{\tau_i^\infty = i} Z^{(\tau_{i+1}^\infty)} + 1_{\tau_i^\infty > i} Y^{\infty(i)} \\ &= Y^{\infty(i)} + 1_{\tau_i^\infty = i} (E^{\mathcal{F}^{(i)}} Z^{(\tau_{i+1}^\infty)} - Y^{\infty(i)}). \end{aligned} \tag{5.23}$$

It is not difficult to see that

$$\tau_i^\infty = \inf\{j : i \le j \le k, \ \max_{p:\, j \le p \le k} E^{\mathcal{F}^{(j)}} Z^{(\tau_p^\infty)} \le Z^{(j)}\}, \quad 0 \le i \le k, \tag{5.24}$$

by letting $m \uparrow \infty$ in (5.19), the definition of $\tau_i^{(m)}$. Indeed, from (5.19) it follows that

$$\begin{aligned} \tau_i^\infty = \lim_{m \uparrow \infty} \tau_i^{(m)} &= \lim_{m \uparrow \infty} \inf\{j : i \le j \le k, \ \widetilde{Y}^{m(j)} \le Z^{(j)}\} \\ &\le \inf\{j : i \le j \le k, \ \widetilde{Y}^{\infty(j)} \le Z^{(j)}\}, \quad 0 \le i \le k, \end{aligned}$$

since $\widetilde{Y}^{m(j)}$ converges to $\widetilde{Y}^{\infty(j)}$, non-decreasingly in $m$. Now suppose that

$$\tau_i^\infty < \inf\{j : i \le j \le k, \ \widetilde{Y}^{\infty(j)} \le Z^{(j)}\}.$$

Then it follows that for $m$ large enough

$$\widetilde{Y}^{\infty(l)} \ge \widetilde{Y}^{m(l)} > Z^{(l)}, \quad i \le l \le \tau_i^\infty,$$

hence, $\tau_i^{(m)} > \tau_i^\infty$, a contradiction with (5.21).

Due to (5.24), for each $j$, $0 \le j \le k$, with $j < \tau_i^\infty$ we have

$$\max_{p:\, j \le p \le k} E^{\mathcal{F}^{(j)}} Z^{(\tau_p^\infty)} > Z^{(j)},$$

and

$$\max_{p:\, \tau_i^\infty \le p \le k} E^{\mathcal{F}^{(\tau_i^\infty)}} Z^{(\tau_p^\infty)} \le Z^{(\tau_i^\infty)}.$$

Then, in particular,

$$1_{\tau_i^\infty = i} E^{\mathcal{F}^{(i)}} Z^{(\tau_{i+1}^\infty)} \le 1_{\tau_i^\infty = i} \max_{p:\, i \le p \le k} E^{\mathcal{F}^{(i)}} Z^{(\tau_p^\infty)} \le 1_{\tau_i^\infty = i} Z^{(i)}, \quad 0 \le i < k.$$

Therefore, from (5.23) we obtain

$$E^{\mathcal{F}^{(i)}} Y^{\infty(i+1)} - Y^{\infty(i)} = 1_{\tau_i^\infty = i}(E^{\mathcal{F}^{(i)}} Z^{(\tau_{i+1}^\infty)} - Y^{\infty(i)}) \\ \leq 1_{\tau_i^\infty = i}(Z^{(i)} - Y^{\infty(i)}) \leq 0, \quad 0 \leq i < k, \tag{5.25}$$

where the last inequality follows from (5.20). So the process $(Y^{\infty(i)})$ is a supermartingale. ∎

In this section the presented procedure for constructing the sequence $(Y^m)_{m\geq 0}$ has some flavour of Picard iteration. We may therefore expect rapid convergence in some sense. In this respect, Kolodko & Schoenmakers (2004) derived the following expression for the distance between two subsequent iterations,

$$Y^{m+1(i)} - Y^{m(i)} = E^{\mathcal{F}^{(i)}} \sum_{p=i}^{\tau_i^{(m+1)}-1} 1_{\tau_p^{(m)}=p}(E^{\mathcal{F}^{(p)}} Y^{m(p+1)} - Y^{m(p)}) \geq 0. \tag{5.26}$$

The proof is given in Appendix A.3.2. Note that trivially $Y^{m(k)} = Z^{(k)}$ is constant for $m \geq 0$. Then, by using backward induction, the following funny result can be derived from (5.26); see Appendix A.3.3 and Bender & Schoenmakers (2004).

***PROPOSITION 5.2***
*For $0 \leq i \leq k$ it holds*

$$m \geq k - i \Longrightarrow Y^{m+1(i)} = Y^{m(i)} = Y^{*(i)} \quad a.s.$$

*As a consequence, after $k$ iterations the Snell envelope is constructed.*

Note that for constructing the whole Snell envelope the backward dynamic program requires $k$ iterations also. However, for a small $m$ (e.g., $m = 2$), iteration $Y^m$ (e.g., $Y^2$) provides an approximation of the *whole* Snell envelope, whereas $m$ steps in the backward dynamic program give only the values $Y^{*(k-m)}, \ldots, Y^{*(k)}$.

Due to the definition of the stopping family $(\tau_i^{(m)})$, we have also the following result.

***COROLLARY 5.1***
*Let $m \geq k - i$. Then,*

$$\tau_i^{(m)} = \inf\{j : i \leq j \leq k,\ Y^{*(j)} \leq Z^{(j)}\},$$

*i.e., $(\tau_i^{(m)})$ coincides with the first optimal stopping time for $m \geq k - i$.*

### 5.4.3 On the Implementation of the Iterative Procedure

For the usual situation where the filtration $(\mathcal{F}^{(i)})$ is generated by a discrete Markov process (e.g., as in Section 5.3), the implementation of the proposed iterative procedure in Section 5.4.2 is straightforward and can be done in a generic way for a variety of (not necessarily financial) optimal stopping problems. Although such an implementation gives rise to nested Monte Carlo simulation, the degree of nesting can be chosen and, in particular, does not explode like in a Monte Carlo implementation of the backward dynamic program (see Section 5.2). Moreover, in Section 5.7 we will see that for Bermudan swaptions practically correct prices can be obtained already with $m = 2$ via a quadratic Monte Carlo simulation.

Let us consider a Markovian Libor process $L$ (e.g., a market model) and assume that its dynamics is given in a numeraire $B$ like in Remark 5.1. For example, $B$ is the terminal bond. Consider a Bermudan Libor derivative for which the exercise dates coincide with the whole tenor structure, and the underlying European options can be priced (quasi-)analytically (a Bermudan swaption for example). Then, Algorithm 5.1 below sketches the iterative procedure in Section 5.4.2 for the case $m = 2$ and initial stopping family $\tau_i^{(0)} \equiv i$. Note that, on a fixed path $\omega$, the set of exercise dates where a policy $\tau$ says "exercise" is characterised by

$$\{\eta : 0 \leq \eta \leq k, \tau_\eta(\omega) = \eta\}. \tag{5.27}$$

***ALGORITHM 5.1***

*Simulate $M$ trajectories $(L^{(m)}(T_j),\ j = 0, \ldots, n-1)$, $m = 1, \ldots, M$, starting at $L(0)\ \ (= L(T_0))$;*

*Along each trajectory $(m)$ do the following:*

$i := 0$;

(A) *Search the first exercise date $\geq i$ where the policy $\tau^{(1)}$ says "exercise"; hence search $\tau_i^{(1)}$ (e.g., in case of a Bermudan swaption we may use European swaption approximation formulas for this search). This date, say $\eta$, is a candidate for $\tau_0^{(2),(m)}$. To decide whether $\eta \approx \tau_0^{(2),(m)}$ or not we do the following:*

*Simulate $M_1$ trajectories $(L^{(m,p)}(T_q),\ q = \eta, \ldots, n-1)$, $p = 1, \ldots, M_1$, with start value $L^{(m)}(T_\eta)$;*

*Along each trajectory $(m, p)$ search all exercise dates $\geq \eta$ where the policy $\tau^{(1)}$ says "exercise" (e.g. for the Bermudan swaption we may use European swaption approximation formulas again). From these dates we can detect easily via (5.27) (an*

*approximation of) the family*
$(\tau_q^{(1),(m,p)},\ q \geq \eta)$ *along the path* $(m,p)$*;*
*Then, for* $q = \eta, \ldots, k$ *compute*
$Dummy[q] := \dfrac{1}{M_1}\displaystyle\sum_{p=1}^{M_1} Z^{(\tau_q^{(1),(m,p)})} \approx E^{L^{(m)}(T_\eta)} Z^{(\tau_q^{(1)})}$*;*
*Next determine*
$Max\_Dummy := \max\limits_{\eta \leq q \leq k} Dummy[q]$
$\approx \max\limits_{\eta \leq q \leq k} E^{L^{(m)}(T_\eta)} Z^{(\tau_q^{(1)})}$*;*
*Check whether* $Z^{(\eta)} \geq Max\_Dummy$*:*
*If yes, set* $\eta^{(m)} := \eta \approx \tau_0^{(2),(m)}$*;*
*If no, do* $i := \eta + 1$*; if* $i < k$ *go to (A) and repeat;*

*We so end with up an* $\eta$ *for which* $\eta^{(m)} := \eta \approx \tau_0^{(2),(m)}$*;*

*Finally compute* $\dfrac{1}{M}\displaystyle\sum_{m=1}^{M} Z^{(\eta^{(m)})} \approx E^{\mathcal{F}^{(0)}} Z^{(\tau_0^{(2)})}$.

The generalisation of Algorithm 5.1 to $m > 2$ is straightforward. Generally, provided that simple expectations $E^{\mathcal{F}^{(i)}} Z^{(j)}$ (Europeans) can be priced analytically, computation of $Y^m$ requires about $O(N^m)$ simulations of the underlying process. For a person who is only interested in the optimal stopping time (for instance, the optimal exercise decision for the buyer of a Bermudan product), the stopping time $\tau^{(m)}$ can be decided at a cost of $O(N^{m-1})$ simulations.

**REMARK 5.2** *(variance reduced Monte Carlo simulation of* $Y^m$*)* We can reduce the number of Monte Carlo simulations for $Y^m$ by using the following variance reduced representation,

$$Y^{m}(i) = E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m)})} = E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m-1)})} + E^{\mathcal{F}^{(i)}} (Z^{(\tau_i^{(m)})} - Z^{(\tau_i^{(m-1)})}). \quad (5.28)$$

One can expect that $Z^{(\tau_i^{(m-1)})}$ and $Z^{(\tau_i^{(m)})}$ are strongly correlated and thus the variance of $(Z^{(\tau_i^{(m)})} - Z^{(\tau_i^{(m-1)})})$ will be less than the variance of $Z^{(\tau_i^{(m)})}$. So, the computation of $E^{\mathcal{F}^{(i)}} (Z^{(\tau_i^{(m)})} - Z^{(\tau_i^{(m-1)})})$ for a given accuracy can usually be done with less Monte Carlo simulations than needed for direct simulation of $E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m)})}$. $\Box$

Finally, we note that the efficiency of the general iterative procedure for solving discrete optimal stopping problems proposed in this section can be improved in combination with several variance reduction techniques, for instance, by using (5.28) in combination with antithetic variables. Even quasi-Monte Carlo methods look promising in this respect. In any case, we expect that, when the producers of microprocessor chips keep "riding the exponential", computation of higher order iterations, so almost exact prices for Bermudan products, will become feasible in the near future.

## 5.5 Duality; From Tight Lower Bounds to Tight Upper Bounds

In fact, the procedures presented in Section 5.3 and Section 5.4 both provide suboptimal stopping rules for the Bermudan problem and consequently lower estimates for the Bermudan value. In this section we discuss the dual approach which enables the construction of tight upper bounds, from tight lower bounds, in some sense.

### 5.5.1 Dual Approach

The dual method for the optimal stopping problem goes back to Davis & Karatzas (1994), but is further developed in Rogers (2001) in the context of Monte Carlo simulation of continuous time American options, and independently in Haugh & Kogan (2001) for Bermudan style derivatives. The dual method for the Bermudan problem is based on the following observation. For any supermartingale $(S^{(j)})$ with $S^{(0)} = 0$ we have

$$\begin{aligned} Y^{*(0)} &= \sup_{\tau \in \{0,\ldots,k\}} E^{\mathcal{F}_0} Z^{(\tau)} \leq \sup_{\tau \in \{0,\ldots,k\}} E^{\mathcal{F}_0} (Z^{(\tau)} - S^{(\tau)}) \\ &\leq E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - S^{(j)}); \end{aligned} \tag{5.29}$$

hence the right-hand side provides a (dual) upper bound for $Y^{*(0)}$.

Rogers (2001) and Haugh & Kogan (2001) show that the equality in (5.29) is attained at the martingale part of the Doob-Meyer decomposition of $Y^*$,

$$M^{Y^*(0)} = 0; \quad M^{Y^*(j)} = \sum_{l=1}^{j} (Y^{*(l)} - E^{\mathcal{F}^{(l-1)}} Y^{*(l)}), \tag{5.30}$$

and also at the shifted Snell envelope process

$$S^{(j)} = Y^{*(j)} - Y^{*(0)}. \tag{5.31}$$

The next lemma provides a somewhat more general class of supermartingales, which turn (5.29) into an equality. Moreover, we show that the equality holds almost sure.

***LEMMA 5.1***

*Let $\mathcal{S}$ be the set of supermartingales $S$ with $S^{(0)} = 0$. Let $S \in \mathcal{S}$ be such that $Z^{(j)} - Y^{*(0)} \leq S^{(j)}$, $1 \leq j \leq k$. Then,*

$$Y^{*(0)} = \max_{0 \leq j \leq k} (Z^{(j)} - S^{(j)}) \quad a.s. \tag{5.32}$$

**PROOF** From the assumptions it follows that $Z^{(j)} - Y^{*(0)} \leq S^{(j)}$, $0 \leq j \leq k$, then, using (5.29),

$$0 \leq E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - S^{(j)}) - Y^{*(0)} = E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - S^{(j)} - Y^{*(0)}) \leq 0.$$

Hence, we have $E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - S^{(j)} - Y^{*(0)}) = 0$ and $Z^{(j)} - S^{(j)} - Y^{*(0)} \leq 0$, which yields (5.32). ∎

Note that both (5.30) and (5.31) satisfy the conditions of Lemma 5.1; for more examples see Kolodko & Schoenmakers (2004).

**REMARK 5.3** Jamshidian (2003,2004) has presented a multiplicative analogue to the dual representation, both in continuous and discrete time. In the discrete setting we have analogue to (5.29),

$$Y^{*(0)} = \inf_{N \in \mathcal{M}} E^{\mathcal{F}_0} \max_{0 \leq j \leq k} Z^{(j)} \frac{N^{(k)}}{N^{(j)}}, \tag{5.33}$$

where $\mathcal{M}$ is the set of positive martingales $(N^{(j)})$, with $N^{(0)} = 1$. The derivation of the multiplicative representation (5.33) is similar to the additive case and, moreover, it is not difficult to show that the infimum in (5.33) is attained at the martingale part of the multiplicative Doob-Meyer decomposition of $Y^*$,

$$N^{Y^*(0)} = 1; \quad N^{Y^*(j)} = \prod_{l=1}^{j} \frac{Y^{*(l)}}{E^{\mathcal{F}^{(l-1)}} Y^{*(l)}}. \tag{5.34}$$

∎

### 5.5.2 Converging Upper Bounds from Lower Bounds

The duality representation in Section 5.5.1 provides a simple way to estimate the Snell envelope from above, by using an approximation of the Snell envelope denoted by $\overline{Y}$. Let $\overline{M}$ be the martingale part of the Doob-Meyer decomposition of $\overline{Y}$, satisfying

$$\begin{aligned}
\overline{M}^{(0)} &= 0; \\
\overline{M}^{(j)} &= \overline{M}^{(j-1)} + \overline{Y}^{(j)} - E^{\mathcal{F}^{(j-1)}}\overline{Y}^{(j)} \\
&= \sum_{l=i+1}^{j} \overline{Y}^{(l)} - \sum_{l=i+1}^{j} E^{\mathcal{F}^{(l-1)}}\overline{Y}^{(l)},\ j = 1, \ldots, k.
\end{aligned}$$

Then, according to (5.29),

$$Y^{*(0)} \leq E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - \overline{M}^{(j)}) =: \overline{Y}_{up}^{(0)}. \tag{5.35}$$

If $\overline{Y}$ is a lower bound process which dominates the discounted cash-flows, then the gap between $\overline{Y}^{(0)}$ and $\overline{Y}_{up}^{(0)}$ depends, in some sense, on how far the lower bound process $\overline{Y}$ is away from being a supermartingale.

***THEOREM 5.3***
*(Kolodko & Schoenmakers (2003,2004a)) Suppose that* $Y^{*(j)} \geq \overline{Y}^{(j)} \geq Z^{(j)}$, $0 \leq j \leq k$. *Then,*

$$0 \leq \overline{Y}_{up}^{(0)} - \overline{Y}^{(0)} \leq E^{\mathcal{F}_0} \sum_{j=0}^{k-1} \max(E^{\mathcal{F}^{(j)}}\overline{Y}^{(j+1)} - \overline{Y}^{(j)},\ 0).$$

**PROOF** By definition (5.35), we have

$$\begin{aligned}
\overline{Y}_{up}^{(0)} &= E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - \sum_{l=1}^{j} \overline{Y}^{(l)} + \sum_{l=1}^{j} E^{\mathcal{F}^{(l-1)}}\overline{Y}^{(l)}) \qquad (5.36) \\
&= \overline{Y}^{(0)} + E^{\mathcal{F}_0} \max_{0 \leq j \leq k} (Z^{(j)} - \overline{Y}^{(j)} + \sum_{l=0}^{j-1} E^{\mathcal{F}^{(l)}}(\overline{Y}^{(l+1)} - \overline{Y}^{(l)})) \\
&=: \overline{Y}^{(0)} + \Delta^{(0)}.
\end{aligned}$$

We thus have the following estimate,

$$\begin{aligned}
\Delta^{(0)} = \overline{Y}_{up}^{(0)} - \overline{Y}^{(0)} &\leq E^{\mathcal{F}_0} \max_{0 \leq j \leq k} \sum_{l=0}^{j-1} (E^{\mathcal{F}^{(l)}}\overline{Y}^{(l+1)} - \overline{Y}^{(l)}) \\
&\leq E^{\mathcal{F}_0} \max_{0 \leq j \leq k} \sum_{l=0}^{j-1} \max(E^{\mathcal{F}^{(l)}}\overline{Y}^{(l+1)} - \overline{Y}^{(l)}, 0) \\
&\leq E^{\mathcal{F}_0} \sum_{j=0}^{k-1} \max(E^{\mathcal{F}^{(j)}}\overline{Y}^{(j+1)} - \overline{Y}^{(j)}, 0).
\end{aligned}$$

□

By the following Theorem 5.4 we can deduce a sequence of upper bound processes convergent to the Snell envelope from a sequence of lower bound processes which converges to the Snell envelope.

**THEOREM 5.4**
*Let us consider a sequence* $(Y^{m})_{m\geq 0}$ *of lower bound processes which dominate the discounted cash-flows, hence* $Y^{m}(j) \leq Y^{*}(j)$ *and* $Y^{m}(j) \geq Z^{(j)}$*, for* $j = 0,\ldots,k$*,* $m \geq 0$*, and which converge to the Snell envelope, i.e.,* $Y^{*}(j) = \lim_{m\to\infty} Y^{m}(j)$*, for* $j = 0,\ldots,k$*. Then, analogue to (5.35) we can construct a sequence of upper bound processes by*

$$\begin{aligned} Y_{up}^{m}(i) &:= E^{\mathcal{F}^{(i)}} \max_{i\leq j\leq k}(Z^{(j)} - \sum_{l=i+1}^{j} Y^{m}(l) + \sum_{l=i+1}^{j} E^{\mathcal{F}^{(l-1)}} Y^{m}(l)) \\ &=: Y^{m}(i) + \Delta^{m}(i), \end{aligned}$$

*and the sequence* $(Y_{up}^{m})_{m\geq 0}$ *converges to the Snell envelope also.*

**PROOF** By Theorem 5.3 it follows that

$$0 \leq \Delta^{m}(i) \leq E^{\mathcal{F}^{(i)}} \sum_{j=i}^{k-1} \max\left(E^{\mathcal{F}^{(j)}} Y^{m}(j+1) - Y^{m}(j),\ 0\right).$$

By letting $m \uparrow \infty$ on the right-hand side, using dominated convergence, and the supermartingale property of $Y^{*}$, it follows that

$$(\text{a.s.}) \lim_{m\to\infty} \Delta^{m}(i) = 0, \quad 0 \leq i \leq k.$$

Hence,

$$(\text{a.s.}) \lim_{m\to\infty} Y_{up}^{m}(i) = (\text{a.s.}) \lim_{m\to\infty} Y^{m}(i) = Y^{*}(i), \quad 0 \leq i \leq k.$$

□

Since we have constructed explicitly a convergent sequence of lower bound processes in Section 5.4, we may conclude this section with the following important corollary.

**COROLLARY 5.2**
*The convergent sequence of lower bounds* $(Y^{m})_{m\geq 0}$*, constructed in Section 5.4, induces via Theorem 5.4 a sequence of upper bounds* $(Y_{up}^{m})_{m\geq 0}$*, which converges to the Snell envelope as well. Moreover, due to Proposition 5.2, the upper bounds coincide with the Snell envelope after finitely many iteration steps as well.*

## 5.6 Monte Carlo Simulation of Upper Bounds

Consider a Bermudan Libor derivative as introduced in Section 5.2 and suppose we have some approximative process $\widetilde{V}_t$ for the price of this Bermudan issued at time $t$. For instance, for some exercise strategy, i.e., a family of integer valued stopping times $\{\tau_t \in \{\kappa(t), ..., k\} : t \geq 0\}$, the process

$$\widetilde{V}_t := B(t)E^{\mathcal{F}_t}\frac{C_{\mathcal{T}_{\tau_t}}}{B(\mathcal{T}_{\tau_t})} = B(t)E^{\mathcal{F}_t}Z^{(\tau_t)} \tag{5.37}$$

is a lower approximation, $\widetilde{V}_t \leq V_t$. However, here $\widetilde{V}_t$ doesn't need to be a strict lower bound, nor a strict upper bound. The discounted process $\widetilde{Y} := \widetilde{V}/B$ can be considered as a $\widetilde{V}$ associated approximation of the Snell envelope process $Y^*$. In the spirit of Section 5.5.2, definition (5.35), we obtain an upper bound for the Bermudan option at $t = 0$ via[2]

$$\frac{V_0}{B(0)} \leq \frac{V_0^{up}}{B(0)} =: E \sup_{1\leq j\leq k}[Z^{(j)} - \sum_{i=1}^{j}\widetilde{Y}^{(i)} + \sum_{i=1}^{j}E^{\mathcal{F}^{(i-1)}}\widetilde{Y}^{(i)}]. \tag{5.38}$$

We now study Monte Carlo estimation of the target upper bound $V_0^{up}$ in(5.38). For this we assume like in Section 5.3 that $L$ is Markovian, and (without loss of generality) that the numeraire $B$ is chosen as in Remark 5.1. For such a numeraire, $Y^{*(i)}$ is obviously a function of $L^{(i)}$ and we therefore assume that in (5.38) $\widetilde{Y}^{(i)}$ is a function of $L^{(i)}$ as well. Note that, for example, for the lower approximation (5.37) this is true. Then, due to Markovianity of $L$, in (5.38) $\mathcal{F}^{(i-1)}$ may be replaced by the $\sigma$-algebra generated by $L^{(i-1)} := L(\mathcal{T}_{i-1})$, for $i = 1, \ldots, k$. So, by general arguments there exist a (regular) conditional probability measure $P(L^{(j)}, \bullet)$, such that for any $\sigma\{L(s), s \geq \mathcal{T}_j\}$-measurable random variable $Z$,

$$[E^{\mathcal{F}^{(j)}}Z](\omega) = [E^{L^{(j)}}Z](\omega) =: \int P(L^{(j)}, d\widetilde{\omega})Z(\widetilde{\omega}) \qquad a.s., \qquad j = 0, \ldots, k$$

(e.g., see Appendix A.1.4). We now consider for each $j$, $j = 1, ..., k$, a sequence of random variables $\left(\xi_i^{(j)}\right)_{i\in\mathbb{N}}$, where for $i \in \mathbb{N}$, $\xi_i^{(j)}$ are i.i.d. copies of $\widetilde{Y}^{(j)}$ under the conditional measure $P(L^{(j-1)}, \bullet)$, independent of the sigma-algebra $\sigma\{L^{(i)} : i = j, \ldots, k\}$. Hence,

$$E^{\mathcal{F}^{(j-1)}}\widetilde{Y}^{(j)} = \int P(L^{(j-1)}, d\widetilde{\omega})\widetilde{Y}^{(j)}(\widetilde{\omega}) = \int P(L^{(j-1)}, d\widetilde{\omega})\xi_i^{(j)}(\widetilde{\omega}), \qquad i \in \mathbb{N}.$$

[2]For minor technical reasons we henceforth assume without restriction that $t = \mathcal{T}_0 = 0$ is excluded from the set of exercise dates.

For a fixed but arbitrary $K \in \mathbb{N}$ we next consider a discrete process $\widetilde{M}^{(K)}$ defined by $\widetilde{M}_0^{(K)} = 0$ and, recursively,

$$\begin{aligned}\widetilde{M}_j^{(K)} &:= \widetilde{M}_{j-1}^{(K)} + \widetilde{Y}^{(j)} - \frac{1}{K}\sum_{i=1}^{K} \xi_i^{(j)} \\ &= \sum_{q=1}^{j} \widetilde{Y}^{(q)} - \sum_{q=1}^{j} \frac{1}{K}\sum_{i=1}^{K} \xi_i^{(q)}, \qquad j = 1, \ldots, k.\end{aligned}$$

The process $\widetilde{M}^{(K)}$ is thus defined on an extended probability space $\Omega \times \prod$ with $\prod := \prod_{j=1}^{k} \mathbb{R}^K$. So a generic sample element in this space is $(\omega, (\xi^{(j)})_{1\le j\le k})$, with $\omega \in \Omega$ being a realisation of the process $L$ and $\xi^{(j)} := (\xi_i^{(j)})_{i=1,\ldots,K} \in \mathbb{R}^K$, for $j = 1, \ldots, k$.

Clearly, $\widetilde{M}^{(K)}$ is a martingale w.r.t. the filtration $(\widetilde{\mathcal{F}}^{(j)})_{j=0,\ldots,k}$, defined by $\widetilde{\mathcal{F}}^{(0)} := \mathcal{F}_0$ and $\widetilde{\mathcal{F}}^{(j)} := \sigma\big\{F \times H : \Omega \supset F \in \mathcal{F}^{(j)}, \prod \supset H \in \sigma\{\xi^{(1)}, \ldots, \xi^{(j)}\}\big\}$, for $j = 1, \ldots, k$, and we observe that

$$\begin{aligned}E \sup_{1\le j\le k}[Z^{(j)} - \widetilde{M}_j^{(K)}] &= EE^{\mathcal{F}^{(k)}} \sup_{1\le j\le k}[Z^{(j)} - \sum_{q=1}^{j}\widetilde{Y}^{(q)} + \sum_{q=1}^{j}\frac{1}{K}\sum_{i=1}^{K}\xi_i^{(q)}] \\ &\ge E \sup_{1\le j\le k}[Z^{(j)} - \sum_{q=1}^{j}\widetilde{Y}^{(q)} + \sum_{q=1}^{j}\frac{1}{K}\sum_{i=1}^{K}E^{\mathcal{F}^{(k)}}\xi_i^{(q)}] \\ &= E \sup_{1\le j\le k}[Z^{(j)} - \sum_{q=1}^{j}\widetilde{Y}^{(q)} + \sum_{q=1}^{j}\frac{1}{K}\sum_{i=1}^{K}E^{\mathcal{F}^{(q-1)}}\xi_i^{(q)}] \\ &= E \sup_{1\le j\le k}[Z^{(j)} - \sum_{q=1}^{j}\widetilde{Y}^{(q)} + \sum_{q=1}^{j}E^{\mathcal{F}^{(q-1)}}\widetilde{Y}^{(q)}] \\ &= \frac{V_0^{up}}{B(0)} \ge \frac{V_0}{B(0)},\end{aligned}$$

where $E^{\mathcal{F}^{(k)}}\xi_i^{(q)} = E^{\mathcal{F}^{(q-1)}}\xi_i^{(q)}$ holds because $\xi_i^{(q)}$ is independent of $L^{(q)}, \ldots, L^{(k)}$. Via the martingale $\widetilde{M}^{(K)}$ we have thus obtained a new upper bound

$$V_0^{up^{up},K} := B(0)E \sup_{1\le j\le k}[Z^{(j)} - \widetilde{M}_j^{(K)}], \tag{5.39}$$

which is larger than our target upper bound $V_0^{up}$.

The upper bound (5.39) is essentially used in the studies of Andersen & Broadie (2001) and Haugh & Kogan (2001). Usually, $V_0^{up^{up},K}$ in (5.39) will be already close to $V_0^{up}$ for numbers $K$ which are much smaller than the number of Monte Carlo trajectories needed for low variance estimation of the mathematical expectation (5.39). Nonetheless, due to the nested simulation required, Monte Carlo simulation of (5.39) is generally not very efficient in practice. In Section 5.8 we study (5.39) further and present new estimation procedures for $V_0^{up}$ which are more efficient than (5.39).

## 5.7 Numerical Evaluation of Bermudan Swaptions by Different Methods

In this section we show some numerical results of the procedures presented in sections 5.3–5.6, applied to Bermudan swaptions. We consider a Libor market model where, as usual, we take for $B$ the spot Libor rolling-over account as numeraire, hence the Libor dynamics (1.22), while taking into account Remark 5.1. The experiments are carried out with the volatility structure,

$$\gamma_i(t) = cg(T_i - t)e_i \quad \text{with} \quad g(s) = g_\infty + (1 - g_\infty + as)e^{-bs}$$

(see (2.25)), and with $e_i$ being $d$-dimensional unit vectors, decomposing some input correlation matrix of rank $d$. For generating Libor models with different numbers of factors $d$, we take as basis a correlation structure of the form

$$\rho_{ij} = \exp(-\varphi|i-j|); \qquad i, j = 1, \ldots, n-1, \tag{5.40}$$

which has full-rank for $\varphi > 0$. Then, for a particular choice of $d$, we deduce from $\rho$ a rank-$d$ correlation matrix $\rho^d$ with decomposition $\rho^d_{ij} = e_i \cdot e_j$, $1 \leq i, j < n$, by principal component analysis (see Section 2.3.5). Of course, instead of (5.40) it is possible to use more general and economically more realistic correlation structures, for instance, the structure (2.46). As model parameters we take a flat 10% initial Libor curve over a 40 period quarterly tenor structure, and further

$$n = 41,\ \delta_i \equiv \delta = 0.25,\ c = 0.2,\ a = 1.5,\ b = 3.5,\ g_\infty = 0.5,\ \varphi = 0.0413. \tag{5.41}$$

The parameter values (5.41) are chosen such that the involved correlation structure and scalar volatilities can be regarded as typical for a Euro or GBP market. For a "practically exact" numerical integration of the SDE (1.22), we use the log-Euler scheme with $\Delta t = \delta/5$ (see, e.g., Chapter 6 and Kurbanmuradov, Sabelfeld and Schoenmakers (2002)).

We now consider Bermudan swaptions for which the exercise dates coincide with the tenor structure between 1 yr. and 10 yr. According to the iterative method introduced in Section 5.4.2, we compute starting from $\tau_i^{(0)} \equiv i$, three successive lower bounds $Y^{m}(0)$: $Y^{0}(0)$, $Y^{1}(0)$ and $Y^{2}(0)$. Further, we compute lower estimations $Y_A^{(0)}$ obtained by Andersen's method outlined in Section 5.3,

**TABLE 5.1:** Comparison of Andersen $Y_A^{(0)}$, Andersen & Broadie $Y_{up,A}^{(0)}$, and the iteration method $Y^{2(0)}$, OTM strike $\theta = 0.12$, base points

| $d$ | $Y^{0(0)}$ (SD) | $Y^{1(0)}$ (SD) | $Y^{2(0)}$ (SD) | $Y_A^{(0)}$ (SD) | $Y_{up,A}^{(0)}$ (SD) |
|---|---|---|---|---|---|
| 1 | 10.2(0.0) | 119.3(0.1) | 131.0(0.9) | 133.5(0.7) | 135.4(0.1) |
| 2 | 5.2(0.0) | 114.5(0.1) | 123.4(0.8) | 119.7(0.7) | 127.4(0.3) |
| 10 | 3.0(0.0) | 104.2(0.1) | 110.5(0.6) | 102.8(0.6) | 113.6(0.3) |
| 40 | 2.7(0.0) | 101.4(0.1) | 106.1(0.6) | 98.8(0.5) | 110.3(0.3) |

and dual upper bounds $Y_{up,A}^{(0)}$, due to the lower bound process $Y_A$, via (5.39), as studied in Andersen & Broadie (2001). The process $Y_A$ has the following form,

$$Y_A^{(i)} := E^{\mathcal{F}^{(i)}} Z^{(\tau_{A,i})}, \quad \text{with} \quad \tau_{A,i} := \inf\left\{ j \geq i : B(T_j) Z^{(j)} > \alpha^{(j)} \right\}, \tag{5.42}$$

where $Z^{(j)}$ are the discounted cash-flows according to (5.10), and the constants $\alpha^{(j)}$ are computed via backward induction; see Section 5.3 for details.

Later, in Section 5.8.3 we will see in the context of a study of efficient upper bound estimators that for 1-factor models the Andersen method due to strategy I gets very close to the Snell envelope. In fact, for one factor the relative distance between $Y_A$ and by $Y_{up,A}$ does not exceed 1.5% in the examples of Section 5.8.3 (see for more examples Andersen & Broadie (2001)). However, in Section 5.8.3 we will see also that, when the number of factors gets larger, this distance increases from ITM to OTM strikes For OTM strikes and more than 2 factors this distance becomes even larger than 10% relative. For this reason we here consider OTM Bermudan swaptions for different numbers of factors, with an OTM strike $\theta = 0.12$. The simulation results for $Y_A^{(0)}$ and $Y_{up,A}^{(0)}$ given in Table 5.1 are taken over from Table 5.3 in Section 5.8.3.

For the iteration $Y^{2(0)}$, we apply the variance reduction technique (5.28). We use 5000000 Monte Carlo trajectories for $Y^{0(0)}$ and $Y^{1(0)}$ and 20 000 Monte Carlo trajectories (with 100 inner simulations) for computation of $Y^{2(0)} - Y^{1(0)}$. As a consequence, the standard deviations of $Y^{2(0)}$ are roughly equal to the corresponding deviations due to the second term in (5.28), and are also comparable with the standard deviations of $Y_A^{(0)}$; see Table 5.1.

From Table 5.1 we see that in all cases, except for the 1-factor case, the secondly iterated lower bound $Y^{2(0)}$ is significantly higher than $Y_A^{(0)}$. Remarkably, for 10 and 40 factors already the first iteration $Y^{1(0)}$ is slightly higher than $Y_A^{(0)}$. In the 1-factor case, where Andersen's lower bound and corresponding dual upper bound are within 1.5% relative, $Y^{2(0)}$ is 1.5% relative below $Y_A^{(0)}$. Note that for more than 1 factor the computed lower bound $Y^{2(0)}$; hence the second iteration can be found more or less in the middle of $Y_A^{(0)}$ and $Y_{A,up}^{(0)}$.

**REMARK 5.4** The construction of the new stopping time $\widehat{\tau}$ from $\tau$ via (5.14) provides a general method for improving any given stopping time $\tau$ with properties (5.11). So in principle we can improve Andersen's process $Y_A$ by constructing $\widehat{Y}_A$ via (5.15). In this respect, we report that computations for two factor OTM cases yielded comparable values for $\widehat{Y}_A$ and $Y^{2(0)}$. ∎

## 5.8 Efficient Monte Carlo Construction of Upper Bounds

As we have seen in Section 5.5, a good stopping rule can be used, on the one hand, to compute a tight lower bound of the Bermudan value, and on the other, to construct a tight upper bound via the duality approach. Unfortunately, however, whereas the lower bound can be computed by a straightforward Monte Carlo simulation, the upper bound construction via (5.39) requires in general a nested (quadratic) Monte Carlo simulation and is therefore not very efficient. In this section we present a method for a more efficient upper bound construction due to Kolodko & Schoenmakers (2003,2004a).

### 5.8.1 Alternative Estimators for the Target Upper Bound

We consider the set up of Section 5.6 again and, as a first step, we now construct a lower bound for the target upper bound $V_0^{up}$ due to some approximative process $\widetilde{V}_t$. Consider an $(\mathcal{F}^{(k)})$-measurable random index $j_{\max}$ which satisfies

$$\sup_{1\le j\le k}[Z^{(j)}-\sum_{q=1}^{j}\widetilde{Y}^{(q)}+\sum_{q=1}^{j}E^{\mathcal{F}^{(q-1)}}\widetilde{Y}^{(q)}] = Z^{(j_{\max})}-\sum_{q=1}^{j_{\max}}\widetilde{Y}^{(q)}+\sum_{q=1}^{j_{\max}}E^{\mathcal{F}^{(q-1)}}\widetilde{Y}^{(q)}.$$

Then, for any integer $K > 0$,

$$\begin{aligned}\frac{V_0^{up}}{B(0)} &= E\big[Z^{(j_{\max})} - \sum_{q=1}^{j_{\max}}\widetilde{Y}^{(q)} + \sum_{q=1}^{j_{\max}}E^{\mathcal{F}^{(q-1)}}\widetilde{Y}^{(q)}\big] \\ &= E\big[Z^{(j_{\max})} - \sum_{q=1}^{j_{\max}}\widetilde{Y}^{(q)} + \sum_{q=1}^{j_{\max}}\frac{1}{K}\sum_{i=1}^{K}E^{\mathcal{F}^{(q-1)}}\xi_i^{(q)}\big] \\ &= E\big[Z^{(j_{\max})} - \sum_{q=1}^{j_{\max}}\widetilde{Y}^{(q)} + \sum_{q=1}^{j_{\max}}\frac{1}{K}\sum_{i=1}^{K}\xi_i^{(q)}\big],\end{aligned}$$

because $E^{\mathcal{F}^{(k)}}\xi_i^{(q)} = E^{\mathcal{F}^{(q-1)}}\xi_i^{(q)} = E^{\mathcal{F}^{(q-1)}}\widetilde{Y}^{(q)}$. This brings us to the idea of localizing $j_{\max}$ for each particular simulation of the process $L$. To this aim, we carry out the following procedure. We consider on the extended probability space $\Omega \times \prod$ the random index $\widehat{j}_{\max}$ which satisfies

$$Z^{(\widehat{j}_{\max})} - \sum_{q=1}^{\widehat{j}_{\max}} \widetilde{Y}^{(q)} + \sum_{q=1}^{\widehat{j}_{\max}} \frac{1}{K} \sum_{i=1}^{K} \xi_i^{(q)} = \sup_{1 \le j \le k} \big[ Z^{(j)} - \sum_{q=1}^{j} \widetilde{Y}^{(q)} + \sum_{q=1}^{j} \frac{1}{K} \sum_{i=1}^{K} \xi_i^{(q)} \big].$$

Next, we extend the probability space once again to $\Omega \times \prod \times \prod$ and simulate *independent copies* $\widehat{\xi}^{(j)} := (\widehat{\xi}_i^{(j)})_{i=1,\dots,K} \in \mathbb{R}^K$, of $\xi^{(j)} \in \mathbb{R}^K$, for $j = 1, \dots, k$. We then consider on $\Omega \times \prod \times \prod$ the random variable

$$Z^{(\widehat{j}_{\max})} - \sum_{q=1}^{\widehat{j}_{\max}} \widetilde{Y}^{(q)} + \sum_{q=1}^{\widehat{j}_{\max}} \frac{1}{K} \sum_{i=1}^{K} \widehat{\xi}_i^{(q)}$$

with expectation

$$\begin{aligned}
\frac{V_0^{up_{low},K}}{B(0)} &:= E\big[ Z^{(\widehat{j}_{\max})} - \sum_{q=1}^{\widehat{j}_{\max}} \widetilde{Y}^{(q)} + \sum_{q=1}^{\widehat{j}_{\max}} \frac{1}{K} \sum_{i=1}^{K} \widehat{\xi}_i^{(q)} \big] \\
&= E\big[ Z^{(\widehat{j}_{\max})} - \sum_{q=1}^{\widehat{j}_{\max}} \widetilde{Y}^{(q)} + \sum_{q=1}^{\widehat{j}_{\max}} \frac{1}{K} \sum_{i=1}^{K} E^{\widetilde{\mathcal{F}}^{(k)}} \widehat{\xi}_i^{(q)} \big] \qquad (5.43) \\
&= E\big[ Z^{(\widehat{j}_{\max})} - \sum_{q=1}^{\widehat{j}_{\max}} \widetilde{Y}^{(q)} + \sum_{q=1}^{\widehat{j}_{\max}} E^{\mathcal{F}^{(q-1)}} \widetilde{Y}^{(q)} \big] \\
&\le E\big[ Z^{(\widehat{j}_{\max})} - \sum_{q=1}^{j_{\max}} \widetilde{Y}^{(q)} + \sum_{q=1}^{j_{\max}} E^{\mathcal{F}^{(q-1)}} \widetilde{Y}^{(q)} \big] = \frac{V_0^{up}}{B(0)},
\end{aligned}$$

where, most importantly, (5.43) holds while the $\widehat{\xi}^q$ are re-sampled *independent* of the determination of $\widehat{j}_{\max}$ and then we have $E^{\widetilde{\mathcal{F}}^{(k)}} \widehat{\xi}_i^{(q)} = E^{\mathcal{F}^{(k)}} \widehat{\xi}_i^{(q)} = E^{\mathcal{F}^{(q-1)}} \widehat{\xi}_i^{(q)} = E^{\mathcal{F}^{(q-1)}} \widetilde{Y}^{(q)}$. As a result, $V_0^{up_{low},K}$ in (5.43) is a lower bound for $V_0^{up}$, for each $K$, $K = 1, 2, \dots$ It is not difficult to show that for $K \to \infty$ both quantities converge to the target.

***PROPOSITION 5.3***

$$V_0^{up^{up},K} \downarrow V_0^{up} \quad \text{and} \quad V_0^{up_{low},K} \uparrow V_0^{up} \quad \text{for} \quad K \to \infty.$$

For a proof see Appendix A.3.4. We so come up with two different Monte Carlo estimators for the target upper bound $V_0^{up}$.

*Lower estimate for $V_0^{up}$:*

$$\widehat{V}_0^{up_{low},K,M} := \frac{B(0)}{M}\sum_{m=1}^{M}\big[Z^{(\widehat{j}_{\max}^{(m)})} - \sum_{q=1}^{\widehat{j}_{\max}^{(m)}}\widetilde{Y}^{(q;m)} + \sum_{q=1}^{\widehat{j}_{\max}^{(m)}}\frac{1}{K}\sum_{i=1}^{K}\widehat{\xi}_i^{(q;m)}\big] \quad (5.44)$$

*Upper estimate for $V_0^{up}$:*

$$\widehat{V}_0^{up^{up},K,M} := \frac{B(0)}{M}\sum_{m=1}^{M}\sup_{1\le j\le k}\big[Z^{(j)} - \sum_{q=1}^{j}\widetilde{Y}^{(q;m)} + \sum_{q=1}^{j}\frac{1}{K}\sum_{i=1}^{K}\xi_i^{(q;m)}\big] \quad (5.45)$$

In (5.44), (5.45), $\widehat{j}_{\max}^{(m)}$ and $\widetilde{Y}^{(q;m)}$ denote the $m$-th independent sample of $\widehat{j}_{\max}$ and $\widetilde{Y}^{(q)}$, respectively. As a third alternative, motivated by Proposition 5.3, the estimators (5.44) and (5.45) can be combined into a convex family of new estimators,

*Combined estimate for $V_0^{up}$:*

$$\widehat{V}_0^{\alpha,K_u,K_l,M} := \alpha\widehat{V}_0^{up^{up},K_u,M} + (1-\alpha)\widehat{V}_0^{up_{low},K_l,M}, \quad (5.46)$$

for $0 < \alpha < 1$, and suitably chosen simulation numbers $K_u, K_l, M$.

In Section 5.8.3 we will demonstrate in practical examples that the combined estimator may have a much higher efficiency than either $\widehat{V}_0^{up^{up},K,M}$ or $\widehat{V}_0^{up_{low},K,M}$.

**Heuristic motivation of the combined estimator**

Let us suppose that, in view of Proposition 5.3, for some $\beta_u, \beta_l > 0$ the following expansions hold,

$$\begin{aligned} V_0^{up^{up},K} &= V_0^{up} + \frac{c_u}{K^{\beta_u}} + o\Big(\frac{1}{K^{\beta_u}}\Big), \quad c_u > 0 \qquad \text{and} \\ V_0^{up_{low},K} &= V_0^{up} - \frac{c_l}{K^{\beta_l}} + o\Big(\frac{1}{K^{\beta_l}}\Big), \quad c_l > 0. \end{aligned} \quad (5.47)$$

Let $\alpha$, $0 < \alpha < 1$, be such that $\alpha c_u - (1-\alpha)c_l = 0$ and let $\kappa_u$, $\kappa_l > 0$ be such that $\kappa_u\beta_u = \kappa_l\beta_l$. Consider for some integer $K$,

$$\widehat{V}_0^{\alpha,[K^{\kappa_u}],[K^{\kappa_l}]} := \alpha V_0^{up^{up},[K^{\kappa_u}]} + (1-\alpha)V_0^{up_{low},[K^{\kappa_l}]} = V_0^{up} + o\Big(\frac{1}{K^{\kappa_u\beta_u}}\Big), \quad (5.48)$$

with brackets denoting the Entier function. We then consider the complexity of the two estimators $\mathcal{U} := \widehat{V}_0^{up^{up},K,M}$ and $\mathcal{A} := \widehat{V}_0^{\alpha,[K^{\kappa_u}],[K^{\kappa_l}],M}$. As usual, the accuracy $\varepsilon$ of an estimator $\widehat{s}$ for a target value $p$ is defined via

$$\varepsilon^2 := E(\widehat{s} - p)^2 = Var(\widehat{s}) + (E\widehat{s} - p)^2$$

and so we may write by (5.47),(5.48),

$$\varepsilon^2_{\mathcal{U}} := \frac{1}{M} Var(\widehat{V}_0^{up^{up},K,1}) + \frac{c_u^2}{K^{2\beta_u}} + o(\frac{1}{K^{2\beta_u}}),$$
$$\varepsilon^2_{\mathcal{A}} := \frac{\alpha^2}{M} Var(\widehat{V}_0^{up^{up},[K^{\kappa_u}],1}) + \frac{(1-\alpha)^2}{M} Var(\widehat{V}_0^{up_{low},[K^{\kappa_l}],1}) + o(\frac{1}{K^{2\kappa_u\beta_u}}),$$

where the simulation of the up-up and up-low estimator is assumed to be done independently.

**REMARK 5.5** In practice it is more efficient to localize $\widehat{j}_{\max}$ using the samples for $V_0^{up^{up},K}$. Then, the up-up and up-low estimator are dependent in general, so

$$\varepsilon^2_{\mathcal{A}} := \frac{1}{M} Var\left(\alpha \widehat{V}_0^{up^{up},[K^{\kappa_u}],1} + (1-\alpha)\widehat{V}_0^{up_{low},[K^{\kappa_l}],1}\right) + o(\frac{1}{K^{2\kappa_u\beta_u}}).$$

$\Box$

Since $Var(\widehat{V}_0^{up^{up},K,1})$ and $Var\left(\alpha \widehat{V}_0^{up^{up},[K^{\kappa_u}],1} + (1-\alpha)\widehat{V}_0^{up_{low},[K^{\kappa_l}],1}\right)$ are uniformly bounded in $K$, we can deduce in the spirit of Schoenmakers & Heemink (1997 ) and Duffy & Glyn (1995) an asymptotically optimal tradeoff between bias and statistical error of the estimators $\mathcal{U}$ and $\mathcal{A}$. In fact, their bias and statistical error should be of comparable magnitude. For the up-up estimator $\mathcal{U}$ we thus take $K \propto M^{1/(2\beta_u)}$ yielding $\varepsilon^2_{\mathcal{U}} \propto M^{-1}$, with $\propto$ denoting asymptotic equivalence, and then for the required computational costs to achieve an accuracy $\varepsilon$ we have

$$\mathrm{Cost}_{\mathcal{U}}(\varepsilon) \propto MK \propto M^{1+\frac{1}{2\beta_u}} \propto \frac{1}{\varepsilon^{2+1/\beta_u}}.$$

For the combined estimator $\mathcal{A}$ we need to know a bit more about the bias term $o(K^{-\kappa_u\beta_u})$ in (5.48). Suppose we can identify a $\gamma > 1$, preferably as large as possible, such that this term may be represented as $O(K^{-\gamma\kappa_u\beta_u})$. Then, by choosing $K \propto M^{1/(2\kappa_u\beta_u\gamma)}$ we obtain in a similar way $\varepsilon^2_{\mathcal{A}} = O(M^{-1})$ and

$$\mathrm{Cost}_{\mathcal{A}}(\varepsilon) \propto MK^{\max(\kappa_u,\kappa_l)} \propto M^{1+\frac{\max(\kappa_u,\kappa_l)}{2\kappa_u\beta_u\gamma}} \propto \frac{1}{\varepsilon^{2+\max(1/\beta_u,1/\beta_l)/\gamma}}.$$

So, under the assumptions above,

$$\frac{\mathrm{Cost}_{\mathcal{U}}(\varepsilon)}{\mathrm{Cost}_{\mathcal{A}}(\varepsilon)} \longrightarrow \infty \quad \text{as} \quad \varepsilon \downarrow 0, \qquad \text{if} \qquad \frac{\beta_u}{\beta_l} < \gamma. \tag{5.49}$$

**REMARK 5.6** We see that the complexity of the combined estimator $\mathcal{A}$ does only depend on the ratio $\kappa_l/\kappa_u = \beta_u/\beta_l$ and thus may take $\kappa_u =$

$\min(1, \beta_l/\beta_u)$ and $\kappa_l = \min(1, \beta_u/\beta_l)$, such that $O(K)$ is always the order of the number of inner simulations. ∎

The above analysis, which is built on some additional assumptions however, explains at a heuristic level why the combined estimator may be superior in several applications; see Section 5.8.3.

### 5.8.2 Two Canonical Approximative Processes

Here we address two canonical lower bound processes for the general Bermudan style derivative which arise from two canonical exercise strategies.

#### Maximum of still alive European options

Suppose the option holder has arrived at a certain exercise date $\mathcal{T}_j$, $1 \leq j \leq k$, and looks at which remaining underlying European instrument has the largest value. More precisely, he considers the stopping index defined by

$$\widetilde{\tau}^{(j)} := \inf\left\{ m \geq j : E^{\mathcal{F}^{(j)}} Z^{(m)} = \max_{j \leq p \leq k} E^{\mathcal{F}^{(j)}} Z^{(p)} \right\}. \tag{5.50}$$

This index is clearly $\mathcal{F}^{(j)}$-measurable and the option holder has the right to pin down his exercise policy at $\mathcal{T}_j$ for whatever reason, by deciding at $\mathcal{T}_j$ to exercise at $\mathcal{T}_{\widetilde{\tau}^{(j)}}$. In fact, this is the same as selling the Bermudan at $\mathcal{T}_j$ as a European option with exercise date $\mathcal{T}_{\widetilde{\tau}^{(j)}}$, thus receiving a cash amount of $\widetilde{Y}^{(j)} B(\mathcal{T}_j)$, with

$$\widetilde{Y}^{(j)}_{\max} := \max_{j \leq p \leq k} E^{\mathcal{F}^{(j)}} Z^{(p)} = E^{\mathcal{F}^{(j)}} Z^{(\widetilde{\tau}^{(j)})} \leq Y^{*(j)}. \tag{5.51}$$

The process $\widetilde{Y}$ in (5.51) is a lower estimation of the Snell envelope $Y$ since the policy (5.50) is suboptimal. For instance, because the optimal policy is not $\mathcal{F}^{(j)}$-measurable.

#### Exercise when cash-flow equals maximum of still alive European options

It is clear that exercising a Bermudan at a time where the cash-flow is below the maximum price of the remaining underlying European options is never optimal. This suggests an alternative exercise strategy defined by the following stopping time,

$$\widehat{\tau}^{(j)} := \inf\left\{ m \geq j : Z^{(m)} \geq \max_{m \leq p \leq k} E^{\mathcal{F}^{(m)}} Z^{(p)} \right\},$$

yielding a lower approximation of the Snell envelope,

$$\widehat{Y}^{(j)}_{\max} := E^{\mathcal{F}^{(j)}} Z^{(\widehat{\tau}^{(j)})} \leq Y^{*(j)}. \tag{5.52}$$

Due to Proposition 5.1 we have

$$\widetilde{Y}_{\max} \leq \widehat{Y}_{\max} \leq Y^{^*};$$

hence the exercise policy $\widehat{\tau}$ is better than $\widetilde{\tau}$. In fact, the process $\widehat{Y}_{\max}$ corresponds to strategy II with $\alpha \equiv 0$ in the Andersen (1999) method; see Section 5.3.

### 5.8.3 Numerical Upper Bounds for Bermudan Swaptions

In this section we investigate numerically the bias of the upper bound estimators (5.44) and (5.45) for different Bermudan swaptions in the Libor market model (1.22). We take the same volatility structure, model parameters, and exercise dates as in Section 5.7, so a flat 10% initial yield curve, parameters (5.41), and three month exercise dates between 1 yr. and 10 yr. As a lower approximation of the Snell envelope process we consider the maximum of still alive swaption process $\widetilde{Y}_{\max}$. Hence, (5.51), where the European option is now a European swaption with discounted cash-flows $Z^{(j)}$ according to (5.10). We further assume that expansion (5.47) holds true, hence,

$$\begin{aligned} V_0^{up^{up},K} - V_0^{up} &= \frac{c_u}{K^{\beta_u}} + o(\frac{1}{K^{\beta_u}}), \\ V_0^{up} - V_0^{up_{low},K} &= \frac{c_l}{K^{\beta_l}} + o(\frac{1}{K^{\beta_l}}), \qquad \beta_u, \beta_l, c_u, c_l > 0, \end{aligned} \tag{5.53}$$

and aim to identify the parameters $\beta_u$, $\beta_l$, $c_u$, $c_l$ in particular cases.

We compute $V_0^{up_{low},K}$ and $V_0^{up^{up},K}$ by estimators (5.44) and (5.45), respectively, with $K = 2^2, 2^3, \ldots, 2^7$ and $M = 30000$, for the examples in Table 5.2. For $M = 30000$ the standard deviations of both estimators are less than 1.5% relative, for all considered $K$. For $K = 128$, the relative distance between $\widehat{V}_0^{up_{low},K,30000}$ and $\widehat{V}_0^{up^{up},K,30000}$ turns out to be within 1.5%, hence the relative standard deviation of both estimators. So we conclude that within a relative accuracy of 1.5% in this sense, both estimators $\widehat{V}_0^{up_{low},128,30000}$ and $\widehat{V}_0^{up^{up},128,30000}$ give a good approximation of the target upper bound $V_0^{up}$. Therefore, we take their average $\widehat{V}_0^{1/2,128,128,30000}$ as an approximation of $V_0^{up}$.

With regard to (5.53) we next determine the coefficients $\beta_u, \beta_l, c_u, c_l$ by linear regression, i.e., we carry out the following minimizations,

$$RMS_u^{rel} = \tag{5.54}$$
$$\sqrt{\sum_{i=2}^{6} \left( \frac{\log(\widehat{V}_0^{up^{up},2^i,30000} - \widehat{V}_0^{1/2,128,128,30000}) - (\log c_u - \beta_u \log 2^i)}{\log(\widehat{V}_0^{up^{up},2^i,30000} - \widehat{V}_0^{1/2,128,128,30000})} \right)^2}$$
$$\longrightarrow \min_{\beta_u, c_u}, \quad \text{and}$$

$$RMS_l^{rel} = \tag{5.55}$$
$$\sqrt{\sum_{i=2}^{6}\left(\frac{\log(\widehat{V}_0^{1/2,128,128,30000} - \widehat{V}_0^{up_{low},2^i,30000}) - (\log c_l - \beta_l \log 2^i)}{\log(\widehat{V}_0^{1/2,128,128,30000} - \widehat{V}_0^{up_{low},2^i,30000})}\right)^2}$$
$$\longrightarrow \min_{\beta_l, c_l},$$

by straightforward differentiating. Note that in the linear regressions (5.54) and (5.55) we exclude terms due to $i = 7$. This is because for $i = 7$ the denominators in (5.54) and (5.55) are basically zero within the considered accuracy. The values of $\beta_u$, $c_u$, $\beta_l$, $c_l$, obtained for different types of swaptions and different number of factors $d$ are given in Table 5.2. We also show in Table 5.2 the "optimal" $\alpha = c_l/(c_u + c_l)$ and ratios $\beta_l/\beta_u = \kappa_u/\kappa_l$.

## Discussion of Numerical Results

According to Table 5.2, the function $\log(V_0^{up^{up},K} - V_0^{up})$ and $\log(V_0^{up_{low},K} - V_0^{up})$ can be approximated rather close by $\log c_u - \beta_u \log K$ and $\log c_l - \beta_l \log K$, respectively, within errors which do not exceed 3.0%. Hence plotting $\log K \rightarrow \log(V_0^{up^{up},K} - V_0^{up})$ and $\log K \rightarrow \log(V_0^{up^{low},K} - V_0^{up})$ gives approximately straight lines. See Figure 5.1 and 5.2 for $d = 40$ (full factor model) and out-of-the-money swaptions with strike $\theta = 12\%$. The values of $\beta_u$ turn out to be roughly equal to one, whereas $\beta_l$ seem to be significantly smaller than one over all. It would be interesting to study this issue in further detail. Then, it is remarkable that the optimal value of $\alpha$ for different strikes and number of factors does not vary too much. The same applies for $\beta_u$ and $\beta_l$ and we therefore consider for all examples the combined upper bound estimator

$$\widehat{V}_0^{0.4,[K^{0.87}],K,M} = 0.4\widehat{V}_0^{up^{up},[K^{0.87}],M} + 0.6\widehat{V}_0^{up_{low},K,M}, \tag{5.56}$$

where $\alpha = 0.4$ is roughly the average value in Table 5.2, and $\kappa_u = 0.87$ and $\kappa_l = 1$ are based on the average of $\beta_l/\beta_u$ and taking into account Remark 5.6. In Figure 5.2 we show for a particular example, strike $\theta = 0.12$ (OTM) and $d = 40$, a plot of the estimator (5.56) together with $\widehat{V}_0^{up^{up},[K^{0.87}],30000}$ and $\widehat{V}_0^{up_{low},K,30000}$ for different values of $K$. Note that even for any $K$ the bias of the combined estimator is negligible within the given accuracy in this example. Later (in Table 5.3) we will see that the bias of the estimator (5.56) for the particular choices of $\alpha, \kappa_u, \kappa_l$, is negligible also for all other examples in Table 5.2, when $K \geq 4$.

We now compare the combined estimator (5.56) with the up-up estimator (5.45) for different strikes and different number of factors $d$. We consider $\widehat{V}_0^{0.4,4,5,90000}$ and $\widehat{V}_0^{up^{up},100,30000}$, where the respective choices of $K$ and $M$ are determined by experiment, such that both the estimations and the (absolute) standard deviations of the estimators are close for different strikes and different number of factors. The results are given in Table 5.3, columns 5,6.

It is easily seen that the combined estimator $\widehat{V}_0^{0.4,4,5,90000}$ is almost 4 times faster than the up-up estimator $\widehat{V}_0^{up^{up},100,30000}$.

**REMARK 5.7** In general, depending on the quality of the Snell-envelope approximation, higher accuracies for dual upper bound estimations may be required and then the efficiency gain of the combined up-low estimator (5.56) with respect to up-up estimator (5.45) can become tremendous in view of (5.49). $\square$

Now we are going to compare the up-up estimations $V_0^{up^{up},K}$, based on the maximum of still alive swaption process (5.51), with up-up estimations due to an approximative lower bound process $\widetilde{Y}_A$, obtained via a particular exercise boundary, constructed by the Andersen (1999) method (see Section 5.3). The process $\widetilde{Y}_A$ is given in (5.42), where the sequence of constants $(\alpha_m)$ is pre-computed by using Andersen strategy I. We here denote the corresponding upper bound, which is studied in Andersen & Broadie (2001), as $V_{0,AB}^{up^{up},K}$.

We compute $\widehat{V}_{0,AB}^{up^{up},100,10000}$ for different strikes and number of factors, and the results are given in Table 5.3, column 4. As we can see, the values of $\widehat{V}_{0,AB}^{up^{up},100,10000}$ and $\widehat{V}_0^{up^{up},100,30000}$ are rather close. In fact, except for the ATM strikes in the one and two factor model, the differences do not exceed 1% relative. For a full factor model and a particular OTM strike we also compare the estimators $\widehat{V}_{0,AB}^{up^{up},K,10000}$ and $\widehat{V}_0^{up^{up},K,30000}$ for different numbers of inner simulations, $K = 1, \ldots, 100$, and conclude that both estimators coincide within the considered accuracy; see Figure 5.2.

In Table 5.3, column 3, we give lower bounds of Bermudan prices $B^*(0)\widetilde{Y}_A^{(0)}$, due to the stopping time $\tau_A^{(0)}$ (see (5.42)). We see that in case of a one factor model the distance between the lower and upper bound of the Bermudan swaption price is rather close for OTM, ATM as well as for ITM strikes. This observation is consistent with the results reported in Andersen & Broadie (2001). For more than one factor however, the gap between lower and upper bound appears to increase. In particular, as we see in Table 5.3, when the number of factors is larger than 1, the gap increases from ITM to OTM strikes and gets even larger than 10% relative for OTM strikes. Apparently, for more factors the exercise region is more complicated in the sense that the exercise region can not be approximated properly by strategy I.

In Table 5.4 we list the required computational time[3] of the up-up upper bound estimators due to Andersen & Broadie and the process given by the maximum of still alive swaptions. We used the Euler scheme with time steps $\Delta t = \delta$ and for practical relevance, we required an accuracy of 1% relative. The cost of pre-computation of the exercise strategy is not taken into account in Table 5.4. In fact, this cost is small compared with the cost of the upper

[3]The simulations are run on a 1 GHz processor

estimators. It appears that for ATM and OTM strikes the upper bound method due to $\widetilde{Y}_{\max}$ is faster than the upper bound method due to $\widetilde{Y}_A$. The main reason is the fact that in the algorithm for simulating $\widetilde{Y}_A$ one needs to construct a Libor trajectory starting at $T_j$ until the exercise condition is fulfilled (for a description of this algorithm see also Andersen & Broadie, (2001)).

Regarding the rather high computation times in Table 5.4, it is clear that an efficiency gain of about a factor 4 (or maybe more), due to application of the combined upper bound estimator, is desirable in practice. Moreover, for a particular Bermudan product we recommend the following procedure.

*Carry out a pre-computation of the optimal $\beta_u, \beta_l$ and $\alpha$ for the given structure, based on up-up and up-low estimations with lower accuracy. Next, take the number $K$ of inner simulations as small as possible and then choose the number $M$ of outer simulations according to the accuracy required.*

For example see Figure 5.3, where the involved parameters $\beta_u, \beta_l$, and $\alpha$ are optimal for the example under consideration and where $K$ can be taken equal to one in fact. We conclude this section with two further remarks.

**REMARK 5.8** Naturally, the numerical analysis based on the maximum of still alive swaption process (5.51) could also be done for the process (5.52). This process is in fact consistent with Andersen strategy II, $\alpha_m \equiv 0$ (Section 5.3). So, on the one hand, this process is dominated from above by a lower bound process due to strategy II with an optimized $(\alpha_m)$. On the other hand, however, as Andersen reports and we found out also, strategy II with optimized $(\alpha_m)$ performs not substantially better than strategy I with optimized $(\alpha_m)$. Therefore, it is expected that the dual upper bound due to process (5.52) will be more or less comparable with the upper bound due to $\widetilde{Y}_A^{(0)}$, which in turn is comparable with the upper bound due to (5.51) for a more than one factor model. Moreover, it is easily seen that the computation of the dual upper bound by the process (5.52) will be more costly. ▯

**REMARK 5.9** It would be interesting to consider the computational aspects of Jamshidian's (2003,2004) multiplicative dual method (5.33) (5.34) also. In particular, since Jamshidian's dual may also require nested simulation, it would be nice to have a similar combined estimator for reducing the number of inner simulations. Further, efficient upper bounds for so called Israeli options, Bermudans, which are cancelable from the issuer's side, would be interesting (see Kühn & Kyprianou (2003)). ▯

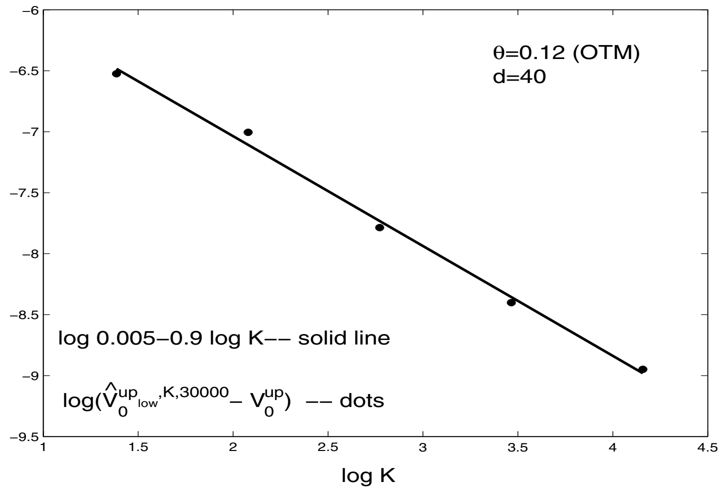


**FIGURE 5.1**: Coefficients in (5.53) for the up-low estimator via logarithmic regression: $\log(\widehat{V}_0^{up_{low},K,30000} - V_0^{up})$ and $\log c_l - \beta_l \log K$ for $\beta_l$ and $c_l$ minimizing (5.55)

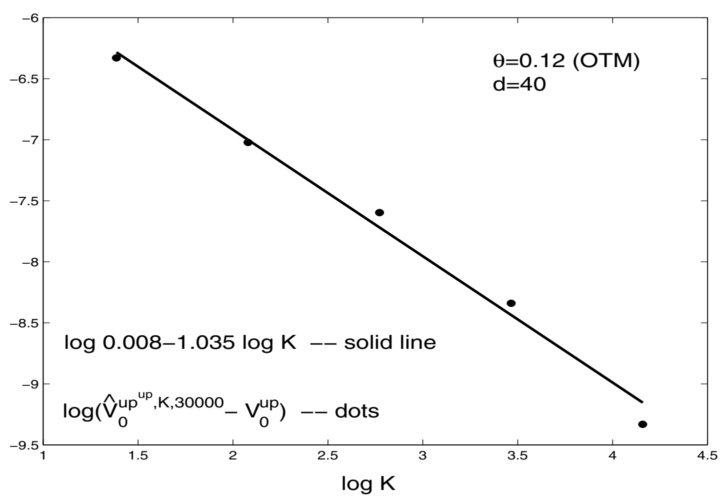


**FIGURE 5.2**: Coefficients in (5.53) for the up-up estimator via logarithmic regression: $\log(\widehat{V}_0^{up^{up},K,30000} - V_0^{up})$ and $\log c_u - \beta_u \log K$ for $\beta_u$ and $c_u$ minimizing (5.54)

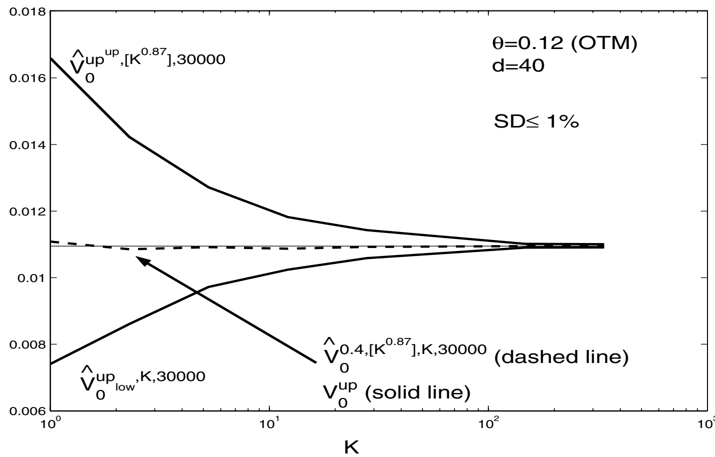


**FIGURE 5.3**: Different Bermudan upper bound estimators due to max of still alive swaptions $\widetilde{Y}_{\max}$

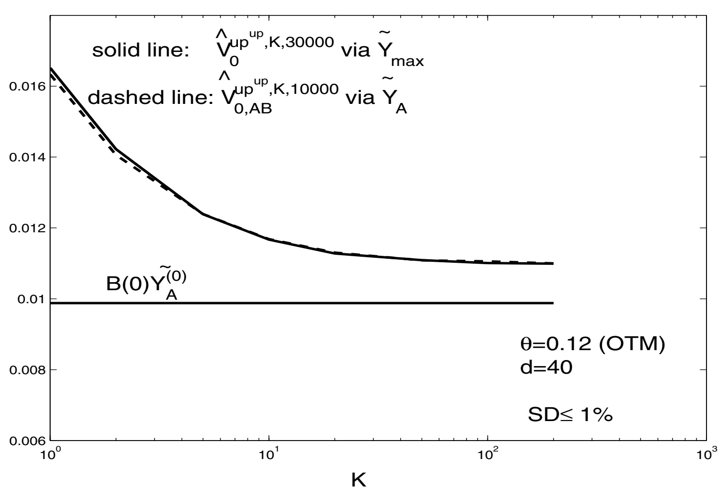


**FIGURE 5.4**: "$up^{up}$" Bermudan upper bound estimators due to $\widetilde{Y}_{\max}$ and $\widetilde{Y}_A$

**TABLE 5.2:** Coefficients of (5.47) for different swaptions and numbers of factors

| $\theta$ | $d$ | $\beta_u$ | $c_u$ | $RMS_u^{rel}$ | $\beta_l$ | $c_l$ | $RMS_l^{rel}$ | $\beta_l/\beta_u$ | $\alpha$ |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1.021 | 0.038 | 0.016 | 0.850 | 0.023 | 0.026 | 0.83 | 0.381 |
| 0.08 | 2 | 0.991 | 0.032 | 0.016 | 0.862 | 0.022 | 0.019 | 0.87 | 0.404 |
| (ITM) | 10 | 0.940 | 0.026 | 0.003 | 0.893 | 0.020 | 0.023 | 0.95 | 0.436 |
| | 1 | 0.970 | 0.025 | 0.021 | 0.746 | 0.013 | 0.015 | 0.77 | 0.335 |
| 0.10 | 2 | 0.872 | 0.020 | 0.009 | 0.840 | 0.014 | 0.029 | 0.96 | 0.417 |
| (ATM) | 10 | 0.968 | 0.021 | 0.016 | 0.717 | 0.010 | 0.020 | 0.74 | 0.317 |
| | 1 | 0.988 | 0.099 | 0.015 | 0.801 | 0.006 | 0.017 | 0.81 | 0.363 |
| 0.12 | 2 | 0.946 | 0.008 | 0.007 | 0.872 | 0.006 | 0.016 | 0.93 | 0.442 |
| (OTM) | 10 | 0.930 | 0.007 | 0.009 | 0.896 | 0.006 | 0.013 | 0.96 | 0.460 |
| | 40 | 1.035 | 0.008 | 0.029 | 0.900 | 0.005 | 0.019 | 0.87 | 0.405 |

**TABLE 5.3:** Upper bounds for different strikes, factors and approximative processes (values in base points)

| $\theta$ | $d$ | $B^*(0)\widetilde{Y}_A^{(0)}$ (SD) | $\widehat{V}_{0,AB}^{up^{up},100,10000}$ (SD) | $\widehat{V}_0^{up^{up},100,30000}$ (SD) | $\widehat{V}_0^{0.4,4,5,90000}$ (SD) |
|---|---|---|---|---|---|
| | 1 | 1116.2(1.6) | 1121.4(0.1) | 1128.8(0.3) | 1124.8(0.4) |
| 0.08 | 2 | 1103.2(1.4) | 1117.6(0.4) | 1121.1(0.3) | 1117.6(0.3) |
| (ITM) | 10 | 1097.1(1.3) | 1111.0(0.4) | 1113.7(0.3) | 1109.5(0.3) |
| | 40 | 1093.2(1.3) | 1106.9(0.4) | 1110.1(0.3) | 1106.9(0.3) |
| | 1 | 403.3(1.2) | 408.3(0.1) | 416.5(0.5) | 416.1(0.5) |
| 0.10 | 2 | 372.6(1.1) | 394.0(0.4) | 397.3(0.5) | 397.7(0.4) |
| (ATM) | 10 | 347.4(1.0) | 373.6(0.5) | 375.8(0.4) | 375.3(0.4) |
| | 40 | 341.6(1.0) | 367.5(0.5) | 368.5(0.4) | 369.8(0.4) |
| | 1 | 133.5(0.7) | 135.4(0.1) | 136.3(0.4) | 135.5(0.3) |
| 0.12 | 2 | 119.7(0.7) | 127.4(0.3) | 127.5(0.3) | 126.5(0.3) |
| (OTM) | 10 | 102.8(0.6) | 113.6(0.3) | 114.5(0.3) | 113.2(0.3) |
| | 40 | 98.8(0.5) | 110.3(0.3) | 109.6(0.3) | 108.6(0.3) |

**TABLE 5.4:** Computation times for upper bounds (in sec.) of the upper estimators in Table 5.3

| $\theta$ | $d$ | $\widehat{V}_{0,AB}^{up^{up},100,1000}$ | $\widehat{V}_0^{up^{up},100,3000}$ |
|---|---|---|---|
| | 1 | 59 | 115 |
| 0.08 | 2 | 69 | 134 |
| (ITM) | 10 | 183 | 241 |
| | 40 | 468 | 603 |
| | 1 | 166 | 113 |
| 0.10 | 2 | 213 | 134 |
| (ATM) | 10 | 510 | 239 |
| | 40 | 1467 | 598 |
| | 1 | 229 | 115 |
| 0.12 | 2 | 299 | 145 |
| (OTM) | 10 | 718 | 263 |
| | 40 | 2076 | 625 |

## 5.9 Multiple Callable Structures

We conclude this chapter with an optional description of Bermudan structures, where the investor has in general more than one exercise right. A typical example is the chooser flex cap which can be regarded as a standard cap with the restriction that the option holder may call only a certain number of caplets during the life time of the cap. In this section we discuss a method of Bender & Schoenmakers (2004), which is a generalisation of the iterative procedure for standard Bermudans (Kolodko & Schoenmakers (2004), Section 5.4) to multiple callable products.

### 5.9.1 The Multiple Stopping Problem

Consider as in Section 5.2 a sequence $(Z(i)\colon\ i = 0, 1, \ldots, k)$ of cash-flows, adapted to some filtration $(\mathcal{F}^{(i)} : 0 \leq i \leq k)$. We now think of an investor who has the right to exercise a cash-flow $D_0$ times under the additional constraint that he has to wait a minimal number of exercise dates $d_0 \in \mathbb{N}$, called *refracting index*, between exercising two rights. The addition of this constraint avoids trivialities, as otherwise the investor would exercise all rights at the same time. For Libor products usually $d_0 = 1$. The investor's problem is now to maximize his expected gain by exercising optimally.

We now formalize the multiple stopping problem. Due to the multiple setting we need additional indexes and therefore we now choose for a slightly different notation of stopping times, cash-flows etc. This change will maintain the readability without causing any confusion with the notations of the previous sections. Let us define $\mathcal{S}_i(D_0, d_0)$ as the set of $\mathcal{F}^{(i)}$ stopping vectors $(\tau_1(i), \ldots, \tau_{D_0}(i))$ such that $i \leq \tau_1(i)$ and, for all $2 \leq j \leq D_0$, $\tau_{j-1}(i) + d_0 \leq \tau_j(i)$. For convenience we use the convention $Z(i) = 0$ for $i \geq k+1$. The multiple stopping problem can now be stated as follows: Find for $0 \leq i \leq k$ a family of stopping vectors $\tau^*(i) \in \mathcal{S}_i(D_0, d_0)$, such that

$$E^{\mathcal{F}^{(i)}} \sum_{j=1}^{D_0} Z(\tau_j^*(i)) = \sup_{\tau \in \mathcal{S}_i(D_0, d_0)} E^{\mathcal{F}^{(i)}} \sum_{j=1}^{D_0} Z(\tau_j) =: Y_{D_0}^*(i).$$

The process $Y_{D_0}^*$ is called the *Snell envelope* of $Z$ under $D_0$ exercise rights. For the simple stopping problem we usually write $Y^*(i) = Y_1^*(i)$.

The multiple stopping problem can be reduced to $D_0$ nested stopping problems with one exercise right in the following way. Consider a sequence of processes $(X_D)_{0 \leq D \leq D_0}$ with $X_0 \equiv 0$ and for $1 \leq D \leq D_0$, $X_D$ being the Snell envelope of the cash-flows $Z(i) + E^{\mathcal{F}^{(i)}} X_{D-1}(i + d_0)$ under one exercise right. In particular, $X_1 = Y_1^* = Y^*$ is the Snell envelope of $Z$. We next define for $1 \leq D \leq D_0$,

$$\sigma_D^*(i) = \inf\{i \leq j \leq k : \; Z(j) + E^{\mathcal{F}^{(j)}} X_{D-1}(j + d_0) \geq X_D(j)\} \tag{5.57}$$

as an optimal stopping family for the simple stopping problem with Snell envelope $X_D$. It is straightforward to show by induction over $D$, that for $1 \leq D \leq D_0$,

$$Y_D^*(i) = X_D(i), \tag{5.58}$$

and that an optimal stopping vector $(\tau_{d,D})_{1 \leq d \leq D}$ for the multiple stopping problem with $D$ exercise rights and cash-flow $Z$ is given by

$$\begin{aligned} \tau_{1,D}^*(i) &= \sigma_D^*(i) \\ \tau_{d+1,D}^*(i) &= \tau_{d,D-1}^*\left(\sigma_D^*(i) + d_0\right) \quad 1 \leq d \leq D-1. \end{aligned} \tag{5.59}$$

The above reduction is intuitively clear and basically says that the investor has to choose the first stopping time of the stopping vector in the following way: Decide, at time $j$, whether it is better to take the cash-flow $Z(j)$ and enter a new contract with $D-1$ exercise rights starting at $j + d_0$, or to keep the $D$ exercise rights. Then, after entering the stopping problem with $D-1$ exercise rights, he proceeds to behave optimally. By this reduction, any algorithm for single optimal stopping problems can, in principle, be applied iteratively to the multiple stopping problem. For example, Carmona & Touzi (2004) suggest to apply backward dynamic programming iteratively to the $D$ stopping problems. However, this leads to even higher nestings of conditional expectations than the dynamic programming approach for the single stopping problem. As an alternative, we here propose to generalize the policy iteration procedure for the simple Bermudan problem in Section 5.4 to an algorithm which simultaneously improves the Snell envelope for $D$ exercise rights, where $D = 1, \ldots, D_0$.

### 5.9.2 Iterative Algorithm for Multiple Bermudan Products

We now explain how the procedure in Section 5.4 can be generalized to the multiple stopping problem, hence the pricing of multiple callable products. Consider a family of Bermudans with $D$ exercise rights for $1 \leq D \leq D_0$ and refracting index $d_0$. Suppose we are given a policy $\sigma_D^{(0)}(i)$, $1 \leq D \leq D_0$, $0 \leq i \leq k$, as an initial guess for an optimal stopping family (5.57). We then may iterate this policy by induction in the following way. Let for $m \geq 0$,

$$\sigma_D^{(m)}(i), \quad 0 \leq i \leq k, \quad 1 \leq D \leq D_0,$$

be the $m$-th improvement upon the originally given family $(\sigma_D^{(0)}(i))_{1 \leq D \leq D_0}$. We may interpret $\sigma_D^{(m)}(i)$ as the exercise date when the investor exercises (according to the $m$-th improvement) the first of his $D$ rights provided he has not exercised prior to time with index $i$. Then

$$\sigma_D^{(m+1)}(i), \quad 0 \leq i \leq k, \quad 1 \leq D \leq D_0,$$

is constructed as follows. First we define a family of stopping vectors $\tau_{1,D}^{(m)}$ by

$$\begin{aligned} &\tau_{1,D}^{(m)}(i) = \sigma_D^{(m)}(i) && (5.60)\\ &\tau_{d+1,D}^{(m)}(i) = \tau_{d,D-1}^{(m)}\left(\sigma_D^{(m)}(i) + d_0\right), \quad 1 \leq d \leq D-1,\ 2 \leq D \leq D_0,\ 0 \leq i \leq k, \end{aligned}$$

where we set $\sigma_D^{(m)}(i) = i$ for $i \geq k+1$, $1 \leq D \leq D_0$ and recall the convention $Z(i) = 0$ for $i \geq k+1$. According to the $m$-th improvement, $\tau_{d,D}^{(m)}(i)$ can be interpreted as the exercise date, when the investor exercises the $d$-th of his $D$ exercise rights, provided he has not exercised his first right prior to time $i$. The $m$-th approximation of the Snell envelope with $D$ exercise rights due to the $m$-th improvement is given by

$$Y_D^{(m)}(i) = E^{\mathcal{F}^{(i)}} \sum_{d=1}^{D} Z(\tau_{d,D}^{(m)}(i)).$$

For constructing a next approximation of the Snell envelope we introduce an intermediate process by

$$\tilde{Y}_D^{(m+1)}(i) = \max_{p \geq i} E^{\mathcal{F}^{(i)}} \sum_{d=1}^{D} Z(\tau_{d,D}^{(m)}(p))$$

and use this as an exercise criterion to obtain a new stopping family,

$$\sigma_D^{(m+1)}(i) = \inf\left\{ j \geq i;\ \ Z(j) + E^{\mathcal{F}^{(j)}}\, Y_{D-1}^{(m)}(j + d_0) \geq \tilde{Y}_D^{(m+1)}(j) \right\},$$

with the convention $Y_0^{(m)}(i) \equiv 0$. Then, via (5.60) we derive a stopping family $\tau_{1,D}^{(m+1)}$ and the $m+1$ approximation of the Snell envelope is given by

$$Y_D^{(m+1)}(i) = E^{\mathcal{F}^{(i)}} \sum_{d=1}^{D} Z(\tau_{d,D}^{(m+1)}(i)).$$

We now prove the following generalisation of Theorem 5.1.

***THEOREM 5.5***
*(Bender & Schoenmakers (2004)) Suppose $\sigma_D^{(0)}(i)$ satisfies property (5.11) for all $1 \le D \le D_0$. Then, for all $m \ge 0$, $1 \le D \le D_0$, and $0 \le i \le k$,*

$$Y_D^{(m+1)}(i) \ge Y_D^{(m)}(i).$$

*Moreover, for $m \ge Dk - i$,*

$$Y_D^{(m)}(i) = Y_D^*(i),$$

*where $Y_D^*(i)$ denotes the Snell envelope of $Z$ under $D$ exercise rights at date $i$.*

**PROOF** We prove the theorem by induction over $D$. For $D = 1$ the algorithm coincides with the procedure described in Section 5.4 for which the results holds. For the step from $D-1$ to $D$ we start with some preliminary considerations. Define a family of auxiliary processes by

$$Z_D^{(m+1)}(i) = Z(i) + E^{\mathcal{F}^{(i)}}\, Y_{D-1}^{(m)}(i + d_0).$$

Then, for $p \ge j$,

$$\begin{aligned} E^{\mathcal{F}^{(j)}}\, Z_D^{(m+1)}(\sigma_D^{(m)}(p)) &= E^{\mathcal{F}^{(j)}} \left[ Z(\sigma_D^{(m)}(p)) + \sum_{d=1}^{D-1} Z(\tau_{d,D-1}^{(m)}(\sigma_D^{(m)}(p) + d_0)) \right] \\ &= E^{\mathcal{F}^{(j)}} \left[ Z(\tau_{1,D}^{(m)}(p)) + \sum_{d=1}^{D-1} Z(\tau_{d+1,D}^{(m)}(p)) \right] \\ &= E^{\mathcal{F}^{(j)}} \left[ \sum_{d=1}^{D} Z(\tau_{d,D}^{(m)}(p)) \right], \end{aligned}$$

making use of the definition of $\tau_{d+1,D}^{(m)}(p)$. In particular,

$$Y_D^{(m)}(i) = E^{\mathcal{F}^{(i)}}\, Z_D^{(m+1)}(\sigma_D^{(m)}(i)) \tag{5.61}$$

$$\tilde{Y}_D^{(m+1)}(j) = \max_{p \ge j} E^{\mathcal{F}^{(j)}}\, Z_D^{(m+1)}(\sigma_D^{(m)}(p)). \tag{5.62}$$

Thus, the definition of $\sigma_D^{(m+1)}(i)$ can be rewritten as

$$\sigma_D^{(m+1)}(i) = \inf \left\{ j \geq i; \;\; Z_D^{(m+1)}(j) \geq \max_{p \geq j} E^{\mathcal{F}^{(j)}} \, Z_D^{(m+1)}(\sigma_D^{(m)}(p)) \right\}. \quad (5.63)$$

By (5.61)–(5.63) the step from $\sigma_D^{(m)}(i)$ to $\sigma_D^{(m+1)}(i)$ is a one step improvement as described in Section 5.4. Thus,

$$Y_D^{(m)}(i) \leq E^{\mathcal{F}^{(i)}} \, Z_D^{(m+1)}(\sigma_D^{(m+1)}(i)).$$

Now, by the induction hypothesis,

$$Z_D^{(m+1)}(i) = Z(i) + E^{\mathcal{F}^{(i)}} \, Y_{D-1}^{(m)}(i+d_0) \leq Z(i) + E^{\mathcal{F}^{(i)}} \, Y_{D-1}^{(m+1)}(i+d_0) = Z_D^{(m+2)}(i).$$

Hence, we may conclude from (5.61),

$$E^{\mathcal{F}^{(i)}} \, Z_D^{(m+1)}(\sigma_D^{(m+1)}(i)) \leq E^{\mathcal{F}^{(i)}} \, Z_D^{(m+2)}(\sigma_D^{(m+1)}(i)) = Y_D^{(m+1)}(i).$$

Therefore, $Y_D^m(i)$ is increasing in $m$.

Now fix $0 \leq i_0 \leq k$ and $m_0 \geq Dk - i_0$. By the induction hypothesis,

$$Y_{D-1}^{(m)}(i) = Y_{D-1}^*(i),$$

for all $m \geq (D-1)k - i$. In particular,

$$Z_D^{(m+1)}(i) = Z(i) + E^{\mathcal{F}^{(i)}} \, Y_{D-1}^{(m)}(i + d_0) = Z(i) + E^{\mathcal{F}^{(i)}} \, Y_{D-1}^*(i + d_0)$$

for all $0 \leq i \leq k$ and $m \geq (D-1)k$. This means that from step $(D-1)k$ we have an iteration procedure as in the case of a single exercise right, but with the cash-flow $Z(i)$ replaced by $Z(i) + E^{\mathcal{F}^{(i)}} \, Y_{D-1}^*(i + d_0)$. The time $i$ value of such an iteration does not change after $k - i$ new improvements by Proposition 5.2, but coincides with the Snell envelope at that time. Hence, for $m_0 \geq Dk - i_0 = (D-1)k + k - i_0$, $Y_D^{(m_0)}(i_0)$ coincides with the time $i_0$ value of the Snell envelope of $Z(i) + E^{\mathcal{F}^{(i)}} \, Y_{D-1}^*(i + d_0)$ with one exercise right, which, in turn, equals $Y_D^*(i_0)$ by (5.58). ∎

We also have an analogue to Corollary 5.1.

***COROLLARY 5.3***

*Let $m \geq Dk - i$. Then,*

$$\sigma_D^{(m)}(i) = \inf \left\{ j : i \leq j \leq k, \; Y_D^*(j) \leq Z(j) + E^{\mathcal{F}^{(j)}} \, Y_{D-1}^*(j + d_0) \right\}.$$

*In particular, the stopping times $(\tau_{d,D}^{(m)}(i) : \; 1 \leq d \leq D)$ are optimal, when $m \geq Dk - i$.*

**PROOF** The representation of $\sigma_D^{(m)}(i)$ follows from Corollary 5.1 taking the second part of the proof of Theorem 5.5 into account. The optimality of $\tau_{d,D}^{(m)}(i)$ is then a simple consequence of (5.58)–(5.59). ∎